

Managing Public Portfolios: Online Appendix

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October 2024

Abstract

This appendix contains the proofs and other details associated with *Managing Public Portfolios*

A Additional details for Section 3

A.1 Proof of Lemma 2(a)

$\bar{\mathcal{R}}_{t+k} = \bar{R}_{t+k}^{rf}$ by Lemma 1, so that the zeroth order approximation of equation (10) implies $\bar{\xi}_{t+k}^{-1} = \frac{\beta M_{t+k+1}}{M_{t+k}} \bar{\xi}_{t+k+1}^{-1} \bar{R}_{t+k+1}^{rf}$ for all $k \geq 0$. The zeroth order approximation of the households' optimality condition (7) gives us $1 = \frac{\beta M_{t+k+1}}{M_{t+k}} \bar{R}_{t+k+1}^{rf}$ for all $k \geq 0$. Combine these two equations to show that $\bar{\xi}_t = \bar{\xi}_{t+k}$ and, therefore, $\bar{\tau}_t = \bar{\tau}_{t+k}$ for all $k \geq 1$.

Multiply equation (10) for period $t+k$ by r_{t+1}^j and take the expectation in period t to get

$$\mathbb{E}_t \frac{1}{\bar{\xi}_{t+k}} r_{t+1}^j = \mathbb{E}_t \frac{\beta M_{t+k+1}}{M_{t+k}} \frac{1}{\bar{\xi}_{t+k+1}} \mathcal{R}_{t+k+1} r_{t+1}^j. \quad (1)$$

Lemma 1 implies that $\mathbb{E}_t \partial_\sigma \mathcal{R}_{t+k+1} \partial_\sigma r_{t+1}^j = \mathbb{E}_t \partial_\sigma R_{t+k+1}^{rf} \partial_\sigma r_{t+1}^j$, so that the second order approximation of (1) is

$$\begin{aligned} & \mathbb{E}_t \partial_\sigma \frac{1}{\bar{\xi}_{t+k}} \partial_\sigma r_{t+1}^j + \frac{1}{2} \frac{1}{\bar{\xi}_{t+k}} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j \\ &= \frac{\beta M_{t+k+1}}{M_{t+k}} \frac{1}{\bar{\xi}_{t+k+1}} \mathbb{E}_t \partial_\sigma R_{t+k+1}^{rf} \partial_\sigma r_{t+1}^j + \frac{\beta M_{t+k+1}}{M_{t+k}} \bar{R}_{t+k+1}^{rf} \mathbb{E}_t \partial_\sigma \frac{1}{\bar{\xi}_{t+k+1}} \partial_\sigma r_{t+1}^j \\ &+ \frac{\bar{R}_{t+k+1}^{rf}}{\bar{\xi}_{t+k+1}} \mathbb{E}_t \partial_\sigma \frac{\beta M_{t+k+1}}{M_{t+k}} \partial_\sigma r_{t+1}^j + \frac{1}{2} \frac{\beta M_{t+k+1}}{M_{t+k}} \frac{\bar{R}_{t+k+1}^{rf}}{\bar{\xi}_{t+k+1}} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j. \end{aligned}$$

From the results obtained in the previous paragraph, we know that $\bar{\xi}_{t+k} = \bar{\xi}_t$ and $\partial_\sigma \frac{1}{\bar{\xi}_{t+k}} = -\frac{\partial_\sigma \tau_{t+k}}{\bar{\xi}'(\bar{\tau}_t)}$ and that $\frac{\beta M_{t+k+1}}{M_{t+k}} \bar{R}_{t+k+1}^{rf} = 1$, so that this equation further simplifies to

$$\begin{aligned} & -\frac{\xi'(\bar{\tau}_t)}{\bar{\xi}(\bar{\tau}_t)} \mathbb{E}_t \partial_\sigma \tau_{t+k} \partial_\sigma r_{t+1}^j + \frac{1}{2} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j = \mathbb{E}_t \partial_\sigma \ln R_{t+k+1}^{rf} \partial_\sigma r_{t+1}^j - \mathbb{E}_t \partial_\sigma \tau_{t+k+1} \partial_\sigma r_{t+1}^j \\ & + \mathbb{E}_t \partial_\sigma \ln \frac{\beta M_{t+k+1}}{M_{t+k}} \partial_\sigma r_{t+1}^j + \frac{1}{2} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j. \end{aligned}$$

Similarly, the household optimality condition (7) implies that $\mathbb{E}_t r_{t+1}^j = \mathbb{E}_t \frac{\beta M_{t+k+1}}{M_{t+k}} R_{t+k+1}^{rf} r_{t+1}^j$, which to the second order of approximation gives

$$\frac{1}{2} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j = \mathbb{E}_t \partial_\sigma \ln R_{t+k+1}^{rf} \partial_\sigma r_{t+1}^j + \mathbb{E}_t \partial_\sigma \ln \frac{\beta M_{t+k+1}}{M_{t+k}} \partial_\sigma r_{t+1}^j + \frac{1}{2} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j.$$

Combine these two equations to show that $\mathbb{E}_t \partial_\sigma \tau_{t+k} \partial_\sigma r_{t+1}^j = \mathbb{E}_t \partial_\sigma \tau_{t+k+1} \partial_\sigma r_{t+1}^j$ for all k, j .

A.2 Proof of Theorem 1

We first consider the zeroth order economy. Using the definition of zero coupon bond prices, we have

$$\bar{Q}_t^k = \frac{\bar{S}_{t+1}}{\bar{S}_t} \times \frac{\bar{S}_{t+2}}{\bar{S}_{t+1}} \times \dots \times \frac{\bar{S}_{t+k}}{\bar{S}_{t+k-1}} = \bar{Q}_t^1 \times \bar{Q}_{t+1}^1 \times \dots \times \bar{Q}_{t+k-1}^1 = \bar{Q}_t^1 \bar{Q}_{t+1}^{k-1}.$$

Furthermore, Lemma 1 implies that excess returns are zero to the zeroth order and $\bar{Q}_{t+1,t+k} = \bar{Q}_{t+1}^{k-1}$. Thus, the budget constraint (4) gives us

$$\bar{B}_t / \bar{Q}_t^1 = \sum_{k=1}^{\infty} \bar{Q}_{t+1}^{k-1} \bar{X}_{t+k} \implies \bar{B}_t = \sum_{k=1}^{\infty} \bar{Q}_t^k \bar{X}_{t+k}, \quad (2)$$

where we used a convention that $\bar{Q}_t^0 = 1$. We have $\bar{X}_{t+k} = \bar{T}_{t+k} - \bar{G}_{t+k}$ and Lemma 2 implies that in the optimum $\bar{T}_{t+k} = \bar{T}_t$ for all k . This allows us to solve for the optimal level of tax revenues, $\bar{T}_t = (\bar{B}_t + \sum_{k=1}^{\infty} \bar{Q}_t^k \bar{G}_{t+k}) / \sum_{k=1}^{\infty} \bar{Q}_t^k$.

Multiply (4) by r_{t+1}^j and take period t expectations to write it as

$$\mathbb{E}_t r_{t+1}^j \sum_{k=1}^{\infty} \mathcal{Q}_{t+1,t+k} X_{t+k} = \mathbb{E}_t r_{t+1}^j \left(R_{t+1}^{rf} + \sum_{i \neq rf} \omega_t^i r_{t+1}^i \right) B_t.$$

The second order approximations of the right hand side and the left hand side of this equation, due to Lemma 1, are

$$RHS \simeq \frac{1}{2} \frac{\bar{B}_t}{\bar{Q}_t^1} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j + \bar{B}_t \sum_{i \neq rf} \bar{\omega}_t^i \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} r_{t+1}^i$$

and

$$LHS \simeq \frac{1}{2} \left(\sum_{k=1}^{\infty} \bar{Q}_{t+1}^{k-1} \bar{X}_{t+k} \right) \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j + \sum_{k=1}^{\infty} \bar{Q}_{t+1}^{k-1} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} X_{t+k} + \sum_{k=1}^{\infty} \bar{Q}_{t+1}^{k-1} \bar{X}_{t+k} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln \mathcal{Q}_{t+1,t+k}.$$

We want to make several observations about these equations. First, the first terms on the right hand sides of these two equations are equal, due to (2). Second, $\mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln \mathcal{Q}_{t+1,t+1} = 0$ since $\mathcal{Q}_{t+1,t+1} = 1$, and $\mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln \mathcal{Q}_{t+1,t+k} = \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln \bar{Q}_{t+1}^{k-1}$ for $k > 1$ due to Lemma 1. Therefore, combining these equations and multiplying both sides by $\bar{Q}_t^1$ we have

$$\begin{aligned} \bar{Q}_t^1 \bar{B}_t \sum_{i \neq rf} \bar{\omega}_t^i \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} r_{t+1}^i &= \sum_{k=1}^{\infty} \bar{Q}_t^{k+1} \bar{X}_{t+k+1} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln \bar{Q}_{t+1}^k \\ &\quad + \sum_{k=1}^{\infty} \bar{Q}_t^k \bar{G}_{t+k} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln G_{t+k} + \sum_{k=1}^{\infty} \bar{Q}_t^k \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} T_{t+k}. \end{aligned} \quad (3)$$

Note that T_{t+k} is only a function of tax rate τ_{t+k} , so we can write $T_{t+k} = T(\tau_{t+k})$ and $\partial_{\sigma} T_{t+k} = T'(\bar{\tau}_{t+k}) \partial_{\sigma} \tau_{t+k}$. But then we have

$$\mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} T_{t+k} = T'(\bar{\tau}_{t+k}) \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \tau_{t+k} = T'(\bar{\tau}_t) \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \tau_{t+1} \quad (4)$$

where the second equality follows from Lemma 2(a). Finally, observe that the last equation in (4) must be equal to zero by Lemma 2(b), which establishes (16).

A.3 Proof of Corollary 1

Observe that in the proof of Theorem 1 upto equation (4) we only used first order properties of the debt level optimality condition¹ (10), and did not use portfolio optimality condition (11) at all. Without invoking this optimality condition, the government budget constraint is

$$\begin{aligned} \bar{Q}_t^1 \bar{B}_t \sum_{i \neq rf} \bar{\omega}_t^i \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i &= \sum_{k=1}^{\infty} \bar{Q}_t^{k+1} \bar{X}_{t+k+1} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \ln Q_{t+1}^k \\ &+ \sum_{k=1}^{\infty} \bar{Q}_t^k \bar{G}_{t+k} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \ln G_{t+k} + T'(\bar{\tau}_t) \left(\sum_{k=1}^{\infty} \bar{Q}_t^k \right) \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \tau_{t+1}. \end{aligned}$$

Using definition of $\bar{\omega}_t^*$, this implies that

$$\sum_{i \neq rf} \left(\bar{\omega}_t^i - \bar{\omega}_t^{*,i} \right) \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i = \frac{T'(\bar{\tau}_t) \left(\sum_{k=1}^{\infty} \bar{Q}_t^k \right)}{\bar{Q}_t^1 \bar{B}_t} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \tau_{t+1}.$$

Taking the second order approximation of (9), we obtain

$$\partial_{\sigma\sigma} \partial_{prfl,j} V = \beta^t \Pr(s^t) \bar{M}_{t+1} \frac{\xi'(\bar{\tau}_{t+1})}{\xi(\bar{\tau}_{t+1})^2} \mathbb{E}_t \partial_\sigma \tau_{t+1} \partial_\sigma r_{t+1}^j.$$

Combining these two expressions, and using zeroth order tax smoothing $\bar{\tau}_t = \bar{\tau}_{t+1}$, we get

$$\sum_{i \neq rf} \left(\bar{\omega}_t^{*,i} - \bar{\omega}_t^i \right) \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i = \underbrace{\frac{T'(\bar{\tau}_t) \xi(\bar{\tau}_t)^2}{-\xi'(\bar{\tau}_t) \bar{B}_t} \frac{\sum_{k=1}^{\infty} \bar{Q}_t^k / \bar{Q}_t^1}{\beta^t \Pr(s^t) \bar{M}_{t+1}}}_{const_t} \partial_{\sigma\sigma} \partial_{prfl,j} V,$$

which is the expression stated in Corollary 1. Note that $const_t > 0$ if $\bar{B}_t > 0$, $-\xi'(\bar{\tau}_t) > 0$, and $T'(\bar{\tau}_t) = \xi(\bar{\tau}_t) \bar{Y}_t > 0$. If v is constant elasticity γ then $\xi(\tau) = 1 - \gamma \frac{\tau}{1-\tau}$ and the peak of the Laffer curve τ^* satisfies $\frac{\tau^*}{1-\tau^*} = \frac{1}{\gamma}$. This implies that if $\tau < \tau^*$ then $\xi(\tau), -\xi'(\tau) > 0$.

B Additional details for Section 4

B.1 Additional details for Section 4.2

First of all, observe that since ξ_t is the transformation of τ_t the proof of Lemma 2 remains unchanged. This implies that $\bar{\tau}_{t+k} = \bar{\tau}_t$ for all t and $\bar{\tau}_t$ is the solution to $\bar{B}_t = \sum_{k=1}^{\infty} \bar{Q}_t^k (\bar{\Theta}_{t+k}(\bar{\tau}_t(1-\bar{\tau}_t)) - \bar{G}_{t+k})$ which is the generalization of equation (2). Let $\bar{T}_{t+k}^{tax} = \bar{\Theta}_{t+k} \bar{\tau}_t (1 - \bar{\tau}_t)$. We have

$$\mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma T_{t+k} = \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma T_{t+k}^{tax} = \bar{T}_{t+k} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \ln \Theta_{t+k} + \underbrace{\bar{\Theta}_{t+k} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \bar{\tau}_t (1 - \bar{\tau}_t)}_{=0},$$

¹This follows since we always pre-multiplied it by r_{t+1}^j prior to taking second order expansions, and $\bar{r}_{t+1}^j = 0$ by Lemma 1.

where the last term is zero following the same steps as in (4). Substitute this equation into (3) to obtain the expression for the optimal portfolio $\bar{w}_t$. If Σ_t matrix is invertible, it can be written as (24).

B.2 Additional details for Section 4.3

In the economy with liquidity premia, the second order approximation of portfolio optimality condition, equation (14), holds but the analogue of equation (15) becomes

$$\frac{1}{2} \frac{\overline{\beta M_{t+1}}}{M_t} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j + \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \frac{\beta M_{t+1}}{M_t} = -\frac{1}{2} \partial_{\sigma\sigma} (w_{t,k} - w_{t,1}).$$

Combining this equation with (14) we obtain

$$\frac{1}{2} \partial_{\sigma\sigma} (w_{t,k} - w_{t,1}) = \frac{\overline{\beta M_{t+1}}}{M_t} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \frac{1}{\xi_{t+1}} = \frac{-\xi'(\bar{\tau}_{t+1})}{\xi(\bar{\tau}_{t+1})} \frac{\overline{\beta M_{t+1}}}{M_t} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \tau_{t+1}.$$

Since to the first order the liquidity premium is zero, conclusions of Lemma 2(a) extend to this economy, which allows us to write the above equation as

$$\mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \tau_{t+1} = \frac{\xi(\bar{\tau}_t)}{-\xi'(\bar{\tau}_t) \bar{Q}_t^1} \frac{1}{2} \partial_{\sigma\sigma} (w_{t,k} - w_{t,1}). \quad (5)$$

Equation (4) still holds but when we combine it with the portfolio optimality condition (5) we obtain

$$\mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} T_{t+k} = T'(\bar{\tau}_t) \frac{\xi(\bar{\tau}_t)^2}{-\xi'(\bar{\tau}_t) \bar{Q}_t^1} \frac{1}{2} \partial_{\sigma\sigma} (w_{t,k} - w_{t,1}) = \frac{\bar{Y}_t \xi(\bar{\tau}_t)^2}{-\xi'(\bar{\tau}_t) \bar{Q}_t^1} \frac{1}{2} \partial_{\sigma\sigma} (w_{t,k} - w_{t,1}).$$

Substitute this equation into (3) to get

$$\begin{aligned} \bar{Q}_t^1 \bar{B}_t \sum_{i \neq rf} \bar{w}_t^i \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} r_{t+1}^i &= \sum_{k=1}^{\infty} \bar{Q}_t^{k+1} \bar{X}_{t+k+1} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln Q_{t+1}^k \\ &+ \sum_{k=1}^{\infty} \bar{Q}_t^k \bar{G}_{t+k} \mathbb{E}_t \partial_{\sigma} r_{t+1}^j \partial_{\sigma} \ln G_{t+k} + \frac{\bar{Y}_t \xi(\bar{\tau}_t)^2 \sum_{k=1}^{\infty} \bar{Q}_t^k}{-\xi'(\bar{\tau}_t) \bar{Q}_t^1} \frac{1}{2} \partial_{\sigma\sigma} (w_{t,k} - w_{t,1}). \end{aligned} \quad (6)$$

This is equation (28) when Σ_t is invertible.

To derive (27), it will be useful to write household optimality conditions (25) in a slightly different form. Consider a perturbation in which households change the quantity of holding a k period bond by an infinitesimal amount until maturity of that bond. The implied optimality condition for that perturbation is

$$M_t Q_t^k = \mathbb{E}_t \beta^k M_{t+k} + M_t Q_t^k w_{t,k} + \mathbb{E}_t M_{t+1} Q_{t+1}^{k-1} w_{t+1,k-1} + \dots + \mathbb{E}_t M_{t+k-1} Q_{t+k-1}^1 w_{t+k-1,1}.$$

The optimality condition for the notional private k period bond is $M_t Q_t^{k,pr} = \mathbb{E}_t \beta^k M_{t+k}$, which implies

$$0 = \left(\frac{1}{Q_t^k} - \frac{1}{Q_t^{k,pr}} \right) \mathbb{E}_t \frac{\beta^k M_{t+k}}{M_t} + w_{t,k} + \mathbb{E}_t \frac{\beta M_{t+1}}{M_t} \frac{Q_{t+1}^{k-1}}{Q_t^k} w_{t+1,k-1} + \dots + \mathbb{E}_t \frac{\beta^{k-1} M_{t+k-1}}{M_t} \frac{Q_{t+k-1}^1}{Q_t^k} w_{t+k-1,1}.$$

Take the second order approximation of this equation and use the fact that to the first order liquidity premia is zero to obtain

$$0 = \partial_{\sigma\sigma} \left(\frac{1}{Q_t^k} - \frac{1}{Q_t^{k,pr}} \right) \overline{Q_t^k} + \partial_{\sigma\sigma} w_{t,k} + \mathbb{E}_t \partial_{\sigma\sigma} w_{t+1,k-1} + \dots + \mathbb{E}_t \partial_{\sigma\sigma} w_{t+k-1,1}.$$

Finally, observe that $\partial_{\sigma\sigma} \left(\frac{1}{Q_t^k} - \frac{1}{Q_t^{k,pr}} \right) \overline{Q_t^k} = -\partial_{\sigma\sigma} \left(\ln Q_t^k - \ln Q_t^{k,pr} \right)$, which implies equation (27).

B.3 Additional details for Section 4.4

Suppose household h has household specific productivity $\theta_{h,t}$ and we partition the households into two groups: $\mathbb{T}$ represent the set of households who can trade bonds and $\mathbb{N}$ represent the set of households who cannot trade bonds. Other than that, we focus on the baseline economy. Individual budget sets are given by

$$c_{h,t+1} + \iota_{h \in \mathbb{T}} \sum_i b_{t+1}^i = (1 - \tau_{t+1}) Y_{h,t+1} + \iota_{h \in \mathbb{T}} \sum_i R_{t+1}^i b_t^i.$$

and optimality implies

$$Y_{h,t} = \theta_{h,t}^{1+\gamma} (1 - \tau_t)^\gamma$$

We can define total output as $Y_t = \sum_h Y_{h,t}$. Assuming a linear tax function, we have

$$\frac{\partial T_t}{\partial \tau_t} = Y_t + \tau_t \sum_h \frac{\partial Y_{h,t}}{\partial \tau_t} = Y_t - \gamma \frac{\tau_t}{1 - \tau_t} \sum_h Y_{h,t} = Y_t \underbrace{\left(1 - \gamma \frac{\tau_t}{1 - \tau_t} \right)}_{\xi_t},$$

so tax revenue elasticity is the same as before.

To compute the welfare effects of the debt and portfolio perturbations, we apply the envelope theorem to each agent h . Changing tax revenues by ϵ will affect household h 's budget by $Y_{h,t+1} \frac{\partial \tau_{t+1}}{\partial T_{t+1}}$. Following the same steps as the representative agent, welfare gain for agent h is given by

$$\partial_{debt} V_h = \beta^t \Pr(s^t) \left[M_{h,t}(s^t) \frac{1}{\xi_t(s^t)} \frac{Y_{h,t}(s^t)}{Y_t(s^t)} - \mathbb{E}_{s^t} \beta M_{t+1} \mathcal{R}_{t+1} \frac{1}{\xi_{t+1}} \frac{Y_{h,t+1}}{Y_{t+1}} \right]$$

$$\partial_{\sigma\sigma}\partial_{prfl,j}V_h = \beta^t \Pr(s^t) \mathbb{E}_t M_{h,t+1} r_{t+1}^j \frac{1}{\xi_{t+1}} \frac{Y_{h,t+1}}{Y_{t+1}}.$$

Combing these for all agents, an optimality condition for the government at $s^t = s^T$ will be

$$\mathbb{E}_t \sum_h \varpi_h M_{h,t+k} (Q_{t+1}^{t-1})^{-1} r_{t+1}^j \frac{Y_{h,t+k}}{Y_{t+k}} \frac{1}{\xi_{t+k}} = 0, \quad (7)$$

where ϖ_h are Pareto weights for household h and $Q_{t,k} \equiv \frac{1}{\mathcal{R}_{t+1}} \times \dots \times \frac{1}{\mathcal{R}_{t+k}}$ is the inverse cumulative return on the government portfolio between periods t and $t+k$. Take second order expansion of (7) to get

$$\begin{aligned} 0 = & \mathbb{E}_t \left\{ \frac{1}{2} \sum_h \varpi_h [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma\sigma} r_{t+1}^j \left[\frac{Y_{h,t+k}}{Y_{t+k}} \frac{1}{\xi_{t+k}} \right] \right. \\ & + \sum_h \varpi_h [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma} \ln(M_{h,t+k}) \partial_{\sigma} r_{t+1}^j \left[\frac{Y_{h,t+k}}{Y_{t+k}} \frac{1}{\xi_{t+k}} \right] \\ & + \sum_h \varpi_h [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma} \ln \left(\frac{Y_{h,t+k}}{Y_{t+k}} \right) \partial_{\sigma} r_{t+1}^j \left[\frac{Y_{h,t+k}}{Y_{t+k}} \frac{1}{\xi_{t+k}} \right] \\ & - \sum_h \varpi_h [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma} \ln(\xi_{t+k}) \partial_{\sigma} r_{t+1}^j \left[\frac{Y_{h,t+k}}{Y_{t+k}} \frac{1}{\xi_{t+k}} \right] \\ & \left. - \sum_h \varpi_h [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma} \ln(Q_{t+1}^{t-1}) \partial_{\sigma} r_{t+1}^j \left[\frac{Y_{h,t+k}}{Y_{t+k}} \frac{1}{\xi_{t+k}} \right] \right\}. \end{aligned}$$

Canceling out the terms that do not depend on h and dividing out by the coefficient on $\mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j$ yields an approximation to the optimality condition (7)

$$\begin{aligned} 0 = & \mathbb{E}_t \left[\frac{1}{2} \partial_{\sigma\sigma} r_{t+1}^j + \sum_h \mu_{h,t+k} \partial_{\sigma} \ln(M_{h,t+k}) \partial_{\sigma} r_{t+1}^j + \partial_{\sigma} \ln(\xi_{t+k}) \partial_{\sigma} r_{t+1}^j + \partial_{\sigma} \ln(Q_{t+1}^{t-1}) \partial_{\sigma} r_{t+1}^j \right. \\ & \left. + \sum_h \mu_{h,t+k} \partial_{\sigma} \ln \left(\frac{Y_{h,t+k}}{Y_{t+k}} \right) \partial_{\sigma} r_{t+1}^j \right] \quad (8) \end{aligned}$$

where $\mu_{h,t+k} \equiv \varpi_h [M_{h,t+k}] \overline{s_{h,t+k}} / (\sum_h \varpi_h [M_{h,t+k}] \overline{s_{h,t+k}})$ are a deterministic sequence of weights that sum to one with $s_{h,t+k} \equiv \frac{Y_{h,t+k}}{Y_{t+k}}$.

As government bonds are perfect substitutes, for all $h \in \mathbb{T}$ we must have

$$\mathbb{E}_t M_{h,t+k} (Q_{t+1}^{t-1})^{-1} r_{t+1}^j = 0.$$

Expanding this equation yields

$$\begin{aligned} 0 = & \frac{1}{2} [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma\sigma} r_{t+1}^j - [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma} \ln(Q_{t+1}^{t-1}) \partial_{\sigma} r_{t+1}^j \\ & + [M_{h,t+k}] \left(Q_{t+1}^{t-1} \right)^{-1} \partial_{\sigma} \ln(M_{h,t+k}) \partial_{\sigma} r_{t+1}^j \end{aligned}$$

for all $h \in \mathbb{T}$. This simplifies to

$$\frac{1}{2} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j = \mathbb{E}_t \left[\partial_{\sigma} \ln (Q_{t+1}^{t-1}) \partial_{\sigma} r_{t+1}^j - \partial_{\sigma} \ln (M_{h,t+k}) \partial_{\sigma} r_{t+1}^j \right] \quad (9)$$

As this holds for all $h \in \mathbb{T}$ we can average over all traders, using weights $\mu_{h,t+k}$, to obtain

$$\frac{1}{2} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j = \mathbb{E}_t \left[\partial_{\sigma} \ln (Q_{t+1,t-1}) \partial_{\sigma} r_{t+1}^j - \partial_{\sigma} \ln (M_{\mathbb{T},t+k}) \partial_{\sigma} r_{t+1}^j \right] \quad (10)$$

where $\ln (M_{\mathbb{T},t+k})$ is the average SDF of all traders:

$$\ln (M_{\mathbb{T},t+k}) \equiv \sum_{h \in \mathbb{T}} \mu_{h,t+k} \ln (M_{h,t+k}) / \sum_{h \in \mathbb{T}} \mu_{h,t+k}.$$

The same equation does not hold for the non-traders but we do have that for all $h \in \mathbb{N}$

$$\begin{aligned} \frac{1}{2} \mathbb{E}_t \partial_{\sigma\sigma} r_{t+1}^j = & \mathbb{E}_t \left[\partial_{\sigma} \ln (Q_{t+1,t-1}) \partial_{\sigma} r_{t+1}^j - \partial_{\sigma} \ln (M_{h,t+k}) \partial_{\sigma} r_{t+1}^j \right. \\ & \left. + (\partial_{\sigma} \ln (M_{h,t+k}) - \partial_{\sigma} \ln (M_{\mathbb{T},t+k})) \partial_{\sigma} r_{t+1}^j \right]. \end{aligned} \quad (11)$$

We can now use equations (9) and (11) substitute for $\frac{1}{2} \partial_{\sigma\sigma} r_{t+1}^j$ in (8) to get

$$\begin{aligned} -\mathbb{E}_t \partial_{\sigma} \ln \xi_{t+k} \partial_{\sigma} r_{t+1}^j = & \mathbb{E}_t \left[\partial_{\sigma} \left\{ \sum_h \mu_{h,t+k} \ln \left(\frac{1}{s_{h,t+k}} \right) \right\} \partial_{\sigma} r_{t+1}^j \right. \\ & \left. + \partial_{\sigma} \left\{ \sum_{h \in \mathbb{N}} \mu_{h,t+k} (\ln (M_{\mathbb{T},t+k}) - \ln (M_{h,t+k})) \right\} \partial_{\sigma} r_{t+1}^j \right]. \end{aligned}$$

We can further simplify this expression by defining

$$\ln (M_{\mathbb{N},t+k}) \equiv \sum_{h \in \mathbb{N}} \mu_{h,t+k} \ln (M_{h,t+k}) / \left(\sum_{h \in \mathbb{N}} \mu_{h,t+k} \right)$$

as the “average” SDF of the non-traders, then

$$\begin{aligned} -\text{cov}_t \left(\ln \xi_{t+k}, r_{t+1}^j \right) \simeq & \text{cov}_t \left(\sum_h \mu_{h,t+k} \ln \left(\frac{1}{s_{h,t+k}} \right), r_{t+1}^j \right) \\ & + \mu_{\mathbb{N},t+k} \text{cov}_t \left(\ln (M_{\mathbb{T},t+k}) - \ln (M_{\mathbb{N},t+k}), \partial_{\sigma} r_{t+1}^j \right) \end{aligned} \quad (12)$$

where $\mu_{\mathbb{N},t+k} \equiv (\sum_{h \in \mathbb{N}} \mu_{h,t+k})$ is the “share” of non-traders. Equation (12) adds two additional terms to optimality equation in Lemma (2) in main text that capture the effect of heterogeneity on the planners desire to smooth taxes. The first term, $\text{cov}_t \left(\sum_h \mu_{h,t+k} \ln \left(\frac{1}{s_{h,t+k}} \right), r_{t+1}^j \right)$, captures the planners desire to raise taxes in states of the world where inequality is high. The second term, $\mu_{\mathbb{N},t+k} \text{cov}_t \left(\ln (M_{\mathbb{T},t+k}) - \ln (M_{\mathbb{N},t+k}), \partial_{\sigma} r_{t+1}^j \right)$, captures the fact that the planner

is trading on behalf of agents without access to asset markets and therefore will want to raise taxes in states of which the non-traders place less weight on relative to those agents with access to asset markets. This effect is scaled by the relative size of the non-traders. Following the steps of Theorem 1 we get (31) where $\Sigma_t^{ineq}[t, k] = cov_k \left(\sum_h \mu_{h,t+k} \ln \left(\frac{1}{s_{h,t+k}} \right), r_{t+1}^j \right)$ is covariance matrix of returns with inequality and $\Sigma_t^M[t, k] = \mu_{\mathbb{N},t+k} cov_t \left(\ln(M_{\mathbb{T},t+k}) - \ln(M_{\mathbb{N},t+k}), r_{t+1}^k \right)$ is the covariance of returns with the relative stochastic discount factors of traders and non-traders with individual weights $\mu_{\mathbb{N},t+k}$ defined above and temporal weights $s_t^{ineq} = \left\{ \frac{Q_t^k \mathbb{E}_t Y_{t+k} \gamma^{-1} (1 - (1+\gamma)\tau_{t+k})^2}{Q_t^1 B_t} \right\}_k$.

B.4 Additional details for Section 4.5

B.4.1 Perturbations

We now derive equations (34) and (35) under the assumption that price responses satisfy (32). As the first step, consider a perturbation that increases issuance of maturity k by ε_k in period t . Using (33), the responses ∂_{ε_k} to such perturbation of taxes are given by

$$\partial_{\varepsilon_k} T_t = -Q_t^k - \Delta_t^k \partial_{\varepsilon_k} Q_t^k, \quad \partial_{\varepsilon_k} T_{t+1} = Q_{t+1}^{k-1} + D_{t+1}^{k-1}, \quad (13)$$

where D_{t+1}^{k-1} is equal to 1 if $k = 1$ and zero otherwise. Household budget constraint written in the quantity form is

$$C_t + \sum_k Q_t^k \tilde{b}_t^k = (1 - \tau_t) Y_t + \sum_k (Q_t^k + D_t^k) \tilde{b}_{t-1}^k.$$

Using the envelope theorem, the welfare impact of this perturbation, up to $\beta^t \Pr(s^t)$, is given by

$$\partial_{\varepsilon_k} V \propto -\frac{M_t}{\xi_t} \partial_{\varepsilon_k} T_t - \mathbb{E}_{t+1} \frac{\beta M_{t+1}}{\xi_{t+1}} \partial_{\varepsilon_k} T_{t+1} - M_t \delta_t^k \partial_{\varepsilon_k} Q_t^k. \quad (14)$$

Combine (13) and (14), using the fact that (32) implies that $\frac{\partial_{\varepsilon_k} Q_t^k}{Q_t^k} = \partial_{\varepsilon_k} \ln Q_t^k = -\varphi'_{k,t}(\tilde{B}_t^k)$ and set $\partial_{\varepsilon_k} V = 0$ to obtain

$$\frac{1}{\xi_t} - \varphi'_{k,t} \left(\frac{1}{\xi_t} \Delta_t^k - \delta_t^k \right) = \mathbb{E}_{t+1} \frac{\beta M_{t+1}}{M_t} \frac{1}{\xi_{t+1}} R_{t+1}^{k-1},$$

where R_{t+1}^0 is the risk-free interest rate.

Using this observations we can construct both the debt level and the portfolio perturbation. The debt level perturbation is equivalent to setting $\varepsilon_k = \omega_t^k \varepsilon / Q_t^k$ for all maturities k , which implies, due to the previous equation, that

$$\frac{1}{\xi_t} - \sum_k \omega_t^k \varphi'_{k,t} \left(\frac{1}{\xi_t} \Delta_t^k - \delta_t^k \right) = \mathbb{E}_{t+1} \frac{\beta M_{t+1}}{M_t} \frac{1}{\xi_{t+1}} \mathcal{R}_{t+1}.$$

The portfolio perturbation is setting $\varepsilon_k = \varepsilon/Q_t^k$ for some maturity k and $\varepsilon_1 = -\varepsilon/Q_t^1$ to obtain, using our assumption $\varphi_{1,t}(\cdot) = 0$,

$$-\omega_t^k \varphi'_{k,t} \left(\frac{1}{\xi_t} \Delta_t^k - \delta_t^k \right) = \mathbb{E}_{t+1} \frac{\beta M_{t+1}}{M_t} \frac{1}{\xi_{t+1}} r_{t+1}^k.$$

If $\delta_t^k/\Delta_t^k = 0$ these two equations reduce to (34) and (35).

Since price effects are zero to the first order, the result of Lemma 2(a) is then unchanged. Lemma 2(b) is obtained by twice differentiating (35) to get

$$\mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \tau_{t+1} = - \frac{\xi(\bar{\tau}_t)}{-\xi'(\bar{\tau}_t) \bar{Q}_t^1} \bar{\Delta}_t^k \varphi'_{k,t}(\tilde{B}_t^k). \quad (15)$$

This is equation (36) written in terms of observables.

B.4.2 Approximation of optimal portfolios

Note that equation (15) has a very similar structure to (5). For this reason, arguments analogous to the proof of equation (6) gives

$$\begin{aligned} \bar{Q}_t^1 \bar{B}_t \sum_{i \neq rf} \bar{\omega}_t^i \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i &= \sum_{k=1}^{\infty} \bar{Q}_t^{k+1} \bar{X}_{t+k+1} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \ln Q_{t+1}^k \\ &+ \sum_{k=1}^{\infty} \bar{Q}_t^k \bar{G}_{t+k} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \ln G_{t+k} - \frac{\bar{Y}_t \xi(\bar{\tau}_t)^2 \sum_{k=1}^{\infty} \bar{Q}_t^k}{-\xi'(\bar{\tau}_t) \bar{Q}_t^1} \bar{\Delta}_t^k \varphi'_{k,t}(\tilde{B}_t^k). \end{aligned} \quad (16)$$

Finally, observe that Δ_t^k can be written as

$$\Delta_t^k = \frac{B_t}{Q_t^k} \left(\frac{Q_t^k \tilde{B}_t^k}{B_t} - \frac{Q_t^k \tilde{B}_{t-1}^{k+1}}{B_t} \right) = \frac{B_t}{Q_t^k} \left(\omega_t^k - \omega_{t-1}^{k,+} \right),$$

where we applied definition of ω_{t-1}^+ given in text. Substitute this into (16) and to obtain expression for the optimal portfolio. If Σ_t is invertible, it can be stated as (38), where ω_t^* is given in (19).

B.4.3 Optimal portfolio without perfectly elastic demand risk-free bond

Optimal portfolio with price effects on all maturities We now extend the analysis to allow for non zero price effects of all maturities including the risk-free bond. The steps are similar to before. The portfolio perturbation now will yield

$$-\frac{1}{\xi_t} \left(\varphi'_{k,t} \Delta_t^k - \varphi'_{1,t} \Delta_t^1 \right) = \mathbb{E}_t \frac{\beta M_{t+1}}{M_t} \frac{1}{\xi_{t+1}} r_{t+1}^k$$

or

$$cov_t \left(r_{t+1}^k, \partial \tau_{t+1} \right) \simeq - \frac{\xi(\tau_t)}{-\xi'(\tau_t) Q_t^1} \left(\Delta_t^k \varphi'_{k,t} - \Delta_t^1 \varphi'_{1,t} \right)$$

and then equation (16) will be

$$\begin{aligned} \sum_{i \neq rf} \bar{\omega}_t^i \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i &= \sum_{k=1}^{\infty} \frac{\bar{Q}_t^{k+1} \bar{X}_{t+k+1}}{\bar{Q}_t^1 \bar{B}_t} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \ln Q_{t+1}^k \\ &+ \sum_{k=1}^{\infty} \frac{\bar{Q}_t^k \bar{G}_{t+k}}{\bar{Q}_t^1 \bar{B}_t} \mathbb{E}_t \partial_\sigma r_{t+1}^j \partial_\sigma \ln G_{t+k} - \frac{\bar{Y}_t \xi(\bar{\tau}_t)^2 \sum_{k=1}^{\infty} \bar{Q}_t^k}{\bar{B}_t - \xi'(\bar{\tau}_t) (\bar{Q}_t^1)^2} \left(\bar{\Delta}_t^k \varphi'_{k,t} - \bar{\Delta}_t^1 \varphi'_{1,t} \right). \end{aligned}$$

We still have

$$\Delta_t^k = \frac{B_t}{Q_t^k} \left(\frac{Q_t^k \tilde{B}_t^k}{B_t} - \frac{Q_t^k \tilde{B}_{t-1}^{k+1}}{B_t} \right) = \frac{B_t}{Q_t^k} \left(\omega_t^k - \omega_{t-1}^{k,+} \right),$$

so the counterpart of formula (38) is

$$\omega_t = \omega_t^* - \Sigma_t^{-1} D_t \left\{ \Lambda_t (\omega_t - \omega_{t-1}^+) - h_t^1 \right\}, \quad (17)$$

where h_t^1 is a vector (same dimension as ω) with elements

$$h_t^1[i] = \frac{Y_t \varphi'_{1,t}}{Q_t^1} (\omega_t^1 - \omega_{t-1}^{01+}) \text{ for all } i, \omega_t^1 = \mathbf{1} - \mathbf{1}^\top \omega_t.$$

Under the assumption that $\varphi'_{k,t}$ are constants denoted by λ_k , we can simplify the expressions further and express the law of motion of ω_t as a linear system.

Define L_t^+ and $L_t^{1,+}$ so that

$$L_t^+ \omega_{t-1} = \omega_{t-1}^+ \quad \omega_{t-1}^{1,+} = L_t^{1,+} \omega_{t-1}.$$

It can be shown that $L_t^+ = \begin{bmatrix} 0 & \frac{B_t Q_t^3}{B_{t+1} Q_t^2} & 0 & \dots \\ 0 & 0 & \frac{B_t Q_t^4}{B_{t+1} Q_t^3} & \dots \\ \dots & \dots & \dots & \dots \end{bmatrix}$ and $L_t^{1,+} = \begin{bmatrix} \frac{B_t Q_t^1}{B_{t+1} Q_t^2} & 0 & 0 & \dots \end{bmatrix}$.

Using this, we can rewrite equation (17) by substituting out h_t^1 as

$$\begin{aligned} \omega_t &= \omega_t^* - \Sigma_t^{-1} D_t \left[\Lambda_t (\omega_t - L_t^+ \omega_{t-1}) - h_t^1 \right] \\ &= \omega_t^* - \Sigma_t^{-1} D_t \Lambda_t \omega_t + \Sigma_t^{-1} D_t \Lambda_t L_t^+ \omega_{t-1} + \Sigma_t^{-1} D_t \lambda_1 \left(\mathbf{1} - \mathbf{1} \mathbf{1}^\top \omega_t - \mathbf{1} L_t^{1,+} \omega_{t-1} \right) \\ &= \left[I + \Sigma_t^{-1} D_t \Lambda_t + \lambda_1 \Sigma_t^{-1} D_t \mathbf{1} \mathbf{1}^\top \right] (\omega_t^* + \Sigma_t^{-1} D_t \lambda_1 \mathbf{1}) \\ &+ \left[I + \lambda_1 \Sigma_t^{-1} D_t \Lambda_t D_t \Sigma_t^{-1} \mathbf{1} \mathbf{1}^\top \right]^{-1} \left[\Sigma_t^{-1} D_t \left(\Lambda_t L_t^+ - \lambda_1 \mathbf{1} L_t^{1,+} \right) \right] \omega_{t-1}. \end{aligned} \quad (18)$$

C Additional details for Section 5

C.1 Nominal Economy

We now describe a nominal version of the benchmark economy. Let P_t be the price level and suppose all securities are nominal. The risk-free bond now refers to a nominal one-period bond that pays one dollar next period. The household and government budget constraint in the nominal economy are

$$P_t C_t + \sum_i b_t^i = (1 - \tau_t) P_t Y_t + \sum_i R_t^i b_{t-1}^i$$

and

$$P_t (T_t - G_t) + \sum_i B_t^i = \sum_i R_t^i B_{t-1}^i,$$

respectively, where $\{b_t^i, B_t^i\}$ are market values in dollars of private and public sector holdings of security i and $R_t^i = \frac{P_t D_t + Q_t^i}{Q_t^i}$ with Q_t^i being the price of security i in dollars is the nominal holding period return on security i . The definition of competitive equilibrium and optimum competitive equilibrium remain unchanged except in nominal economy they are defined for $\{G_t, P_t, S_t\}$.

It is easy to see that the debt perturbation we considered in Section 3 require tax adjustments $\frac{\varepsilon}{P_t(s^t)}$ in s^t and $\frac{\mathcal{R}_{t+1}(s^{t+1})\varepsilon}{P_{t+1}(s^{t+1})}$ in all $s^{t+1} \succeq s^t$ so that equation (8) remains unchanged with the interpretation that $\mathcal{R}_{t+1}(s^{t+1})$ is the nominal return on the government portfolio. A similar argument shows that the portfolio perturbation requires $\frac{r_{t+1}^j(s^{t+1})\varepsilon}{P_{t+1}(s^{t+1})}$ as the tax adjustment and equation (9) remains unchanged too. This means that the proof of Lemma 2 can be extended to the nominal economy with the only change that r_{t+1}^j is the excess nominal return on security i .

We can rewrite equation (4) for the nominal economy as

$$\mathbb{E}_{t+1} \sum_{k=1}^{\infty} \mathcal{Q}_{t+1,t+k} (P_{t+k} T_{t+k} - P_{t+k} G_{t+k}) = (R_{t+1}^{rf} + \sum_{i \neq rf} \omega_t^i r_{t+1}^i) B_t,$$

and applying the same steps as in the proof of Theorem 1, we get that the optimal portfolio satisfies

$$\sum_{i \neq rf} \bar{\omega}_t^i \mathbb{E}_t \partial_{\sigma} r_{t+1}^i \partial_{\sigma} r_{t+1}^j = \sum_{k=1}^{\infty} \frac{\bar{Q}_t^{k+1} \bar{X}_{t+k+1}^{\$}}{\bar{Q}_t^1 \bar{B}_t} \mathbb{E}_t \partial_{\sigma} \ln Q_{t+1}^k \partial_{\sigma} r_{t+1}^j - \sum_{k=1}^{\infty} \frac{\bar{Q}_t^k \bar{G}_{t+k}^{\$}}{\bar{Q}_t^1 \bar{B}_t} \mathbb{E}_t \partial_{\sigma} \ln G_{t+k}^{\$} \partial_{\sigma} r_{t+1}^j, \quad (19)$$

where $\bar{G}_{t+k}^{\$} = \overline{P_{t+k} G_{t+k}}$, $\bar{X}_{t+k}^{\$} = \bar{T}_t^{\$} - \bar{G}_{t+k}^{\$}$ and $\bar{T}_t^{\$} = \frac{\bar{B}_t + \sum_{k=1}^{\infty} \bar{Q}_t^k \bar{G}_{t+k}^{\$}}{\sum_{k=1}^{\infty} \bar{Q}_t^k}$ with $\{B_t\}$ is market value of public portfolio in dollars, Q_{t+1}^k is the nominal price of a hypothetical k period zero

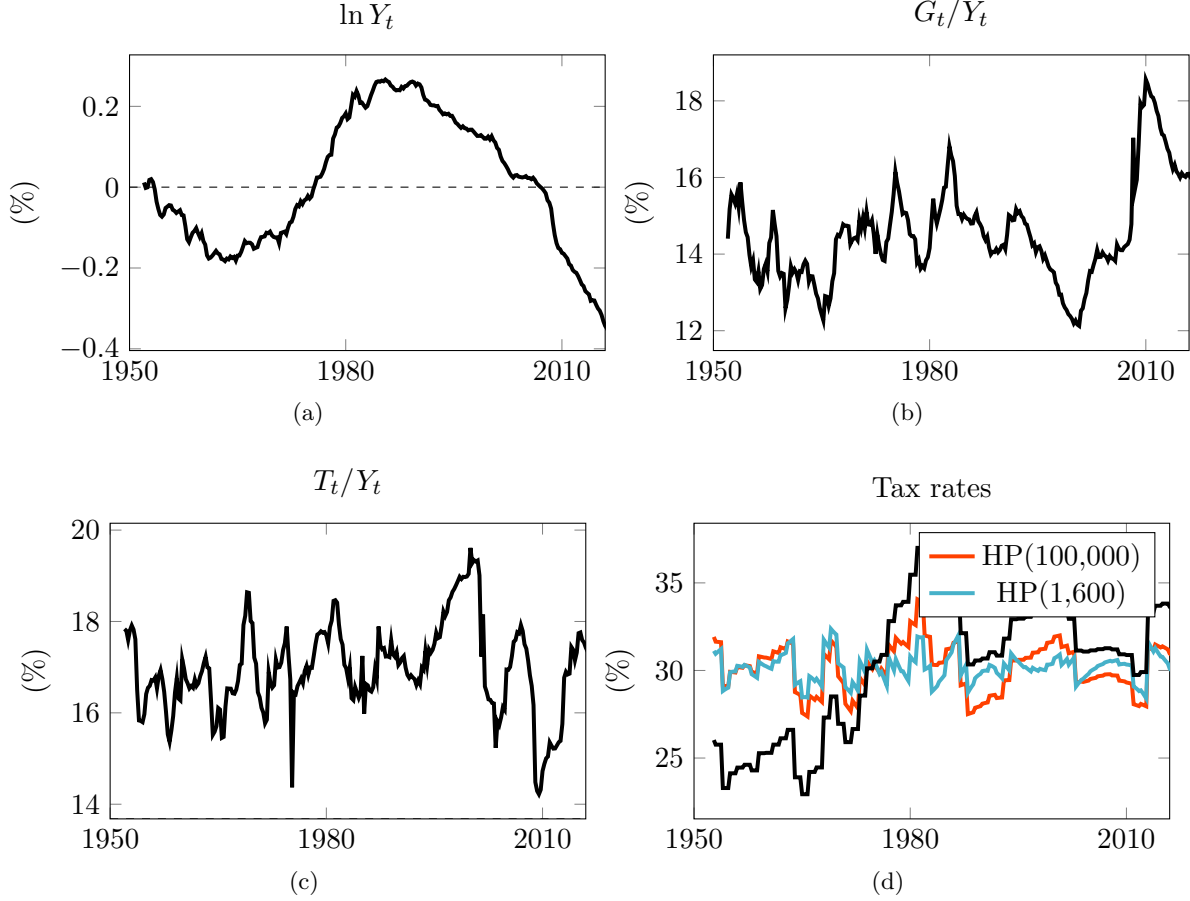


Figure 1: Summary of macroeconomic time series. Panel (a) plots detrended log nominal GDP, panel (b) plots the nominal government expenditure measured as Federal Consumption Expenditures + Federal Transfer Payments To Persons divided by nominal GDP, panel (c) plots nominal revenues divided by nominal GDP, panel (d) plots the average marginal tax rate on income and two ways of detrending the series.

coupon bond in dollars, and $\{r_{t+1}^j\}$ are nominal excess returns on security j . Thus the formula is same as long as we use the appropriate nominal versions of the objects. We drop the \$ superscripts in the main text.

C.2 Data

Output, expenditures, tax revenues

We use the U.S. national income and product accounts to measure output, tax revenues. For our measure of output Y_t we use U.S. GDP. We measure nominal tax revenues T_t as Federal Total Current Tax Receipts + Federal Contribution To Social Insurance and public expenditures G_t as Federal Consumption Expenditures + Federal Transfer Payments To Persons from BEA. All series are nominal and de-trended with constant time trends.

Tax rates

As a measure of tax rates τ_t we use the measure of the average marginal federal tax rate from Barro and Redlick (2011). Their series end in 2012 but we follow their steps and extrapolate this series for the years 2013-2017 using the Statistics of Income publicly available data from the Taxstats website. The series for the raw tax rates are plotted in Figure 1(e). It is clear from the series that there is a structural break in taxes around 1975. In our analysis we want to focus on movements in taxes around business cycle frequencies and therefore we want to remove this break. We pursue two ways of doing that. First, we follow the business cycle literature and apply a Hodrick-Prescott (HP) filter with the penalty parameter set to 1,600. The resulting series is shown as the teal-blue line in the right panel Figure 1(f). While this procedure eliminates the low frequency movements in taxes, it also makes the resulting series “too smooth” post 1975. As an alternative, we adjust the penalty parameter until we achieve both goals: remove low frequency movements around 1975 and preserve the volatility of tax rates after and before 1975. The resulting series is shown in the red line (at a penalty parameter of 100,000) in the right panel. We use the red line as a baseline measure of tax rates, but all our results are virtually unchanged if we use the teal line instead.

Asset returns and government portfolio of bonds

We use the Fama Maturity Portfolios published by CRSP. There are 11 such portfolios, out of which ten portfolios correspond to maturities of 2 to 20 quarter in 2 quarter intervals, and a final portfolio for maturities between 30 and 40 quarters. We use the convention that the upper cut-off for each maturity corresponds to j in the mapping of data to the theory. That is, we use returns on portfolio of bonds of maturities between 2 to 4 quarter to measure r_t^j to $j = 4$, between 4 to 6 quarters to measure r_t^j for $j = 6$, etc. With this convention $j = 40$ is the largest maturity. We aggregate monthly log-returns by summing them across months within each quarter.

To compute the price curve we use yield data from Gürkaynak et al. (2007). The raw data has daily yields for zero coupon bonds of maturities 4, 8, $\dots$ 120 quarters. We interpolate the daily yields using a cubic spline to infer yields for all quarters less than 120. The price curve is computed by using the expression $Q^n = \exp \{-n \times y\}$

Maturity structure of the U.S. government debt

We use the CRSP Treasuries Monthly Series to get the amount outstanding B_t^i for all (including TIPS and other inflation-protected bonds) federally issued (marketable) debt between 1952 and

2017, normalized by its face value. Each bond is uniquely identified by its cusips number n . CRSP also supplies us the Macaulay duration i for the outstanding amount, and the nominal market price $Q_t^{n,i}$ of each bond outstanding. For a few bonds where duration is absent, we set the duration equal to maturity date – current date.

We follow Jiang et al. (2019), and construct at each date t , the market value $Q_t^i B_t^i$ held by the US government in bonds of Macaulay duration i , by summing across cusips n , such that $Q_t^i B_t^i = \sum_n Q_t^{n,i} B_t^{n,i}$. We then sum across all Macaulay duration i to get the market value of the government debt portfolio $B_t \equiv \sum_{i \in \mathcal{G}_t} Q_t^i B_t^i$ at each date t . We finally compute the portfolio weight in the US government debt portfolio for each maturity i using that $\omega_t^i = \frac{Q_t^i B_t^i}{B_t}$.

C.3 Derivations for the baseline factor model

In this section we derive expressions (40) and (41).

From Theorem 1, and discussion of equation (24), the covariances Σ_t , Σ_t^Q and Σ_t^G have elements $\{cov_t(r_{t+1}^i, r_{t+1}^j)\}_{i,j}$, $\{cov_t(\ln Q_{t+1}^k, r_{t+1}^j)\}_{j,k}$, $\{cov_t(\ln G_{t+k}, r_{t+1}^j)\}_{j,k}$, $\{cov_t(\ln \Theta_{t+k}, r_{t+1}^j)\}_{j,k}$ for all j and $k \geq 1$, and weights s_t^Q and s_t^G , s_t^Θ are vectors with elements $\{\frac{Q_t^{k+1} \mathbb{E}_t X_{t+k+1}}{Q_t^1 B_t}\}_k$ and $\{\frac{-Q_t^k \mathbb{E}_t G_{t+k}}{Q_t^1 B_t}\}_k$, $\{\frac{-Q_t^k \mathbb{E}_t T_{t+k}^{tax}}{Q_t^1 B_t}\}_k$. In the baseline case, our factor structure boils down to

$$r_t^k = \kappa_k f_t + \sigma_k \varepsilon_t^k$$

$$\ln G_t = t\Gamma + \kappa_G f_t + \sigma_G \varepsilon_t^G$$

$$\ln \Theta_t = t\Gamma + \kappa_\Theta f_t + \sigma_\Theta \varepsilon_t^\Theta$$

Under this factor structure, we can calculate $(\Sigma_t, \Sigma_t^Q, \Sigma_t^G)$ and $(s_t^Q, s_t^G, s_t^\Theta)$ explicitly. The elements of Σ_t satisfy

$$\begin{aligned} \Sigma_t[j, k] &= cov_t(r_{t+1}^j, r_{t+1}^k) \text{ for } j, k \in \mathcal{G} \\ &= \kappa_j \kappa_k \sigma_f^2 + \iota_{\{j=k\}} \sigma_j^2 \text{ for } j, k \in \mathcal{G} \end{aligned}$$

Lemma 1 implies that $Q_t^1 cov_t(r_{t+1}^k, r_{t+1}^j) \simeq cov_t(\ln Q_{t+1}^k, r_{t+1}^j)$ and we get

$$\Sigma_t^Q[j, k] = Q_t^1 (\kappa_j \kappa_k \sigma_f^2 + \iota_{\{j=k\}} \sigma_j^2) \text{ for } j \in \mathcal{G}, k \in \mathbb{N}$$

The steps to compute the covariances of spending and tax revenues with returns next period are similar, so we show the derivation for just one of them. The elements of Σ_t^G and Σ_t^Θ for $j \in \mathcal{G}, k \in \mathbb{N}$

$$\begin{aligned} \Sigma_t^G[j, k] &= cov_t(\ln G_{t+k}, r_{t+1}^j) \\ &= \kappa_G \kappa_j \sigma_f^2 \end{aligned}$$

$$\Sigma_t^\Theta[j, k] = \kappa_\Theta \kappa_j \sigma_f^2$$

$$\Sigma_T = \Delta + \kappa \Delta_f \kappa^\top$$

Using the Woodbury matrix identity, Σ_t^{-1} can be explicitly computed as

$$\Sigma_t^{-1}[j, k] = \iota_{\{k=j\}} \frac{1}{\sigma_j^2} - \kappa_j \kappa_k \frac{\chi^2}{\sigma_j^2 \sigma_k^2} \text{ for } j, k \in \mathcal{G}$$

where the constant $\chi^{-2} = \sigma_f^{-2} + \sum_{k \in \mathcal{G}} \kappa_k^2 \sigma_k^{-2}$.

We next derive the weights $(s_t^Q, s_t^G, s_t^\Theta)$. The weights on interest rate risk are given by

$$s_t^Q[k] = \frac{Q_t^{k+1} \mathbb{E}_t X_{t+k+1}}{Q_t^1 B_t} = \frac{Q_t^{k+1} \Gamma^{k+1}}{Q_t^1 \sum_{k=1}^\infty Q_t^k \Gamma^k}$$

and the s_t^G, s_t^Θ are

$$s_t^G[k] = \frac{-Q_t^k \mathbb{E}_t G_{t+k}}{Q_t^1 B_t} = \frac{-Q_t^k G_t \Gamma^k}{Q_t^1 B_t}, \quad s_t^\Theta[k] = \frac{Q_t^k \mathbb{E}_t T_{t+k}^{tax}}{Q_t^1 B_t} = \frac{Q_t^k T_t \Gamma^k}{Q_t^1 B_t}$$

Now let us derive the expressions for each of the terms in the portfolio. We start with $\Sigma_t^{-1} \Sigma_t^G s_t^G$ which is the portfolio that hedges the spending risk.

$$\begin{aligned} \Sigma_t^G s_t^G[j] &= - \sum_{k=1}^\infty \kappa_G \kappa_j \sigma_f^2 \left(\frac{Q_t^k G_t \Gamma^k}{Q_t^1 B_t} \right) \\ &\quad - \kappa_G \kappa_j \sigma_f^2 \frac{G_t}{Q_t^1 B_t} \sum_{k=1}^\infty Q_t^k \Gamma^k \end{aligned}$$

Now we multiply with Σ_t^{-1} to get

$$\Sigma_t^{-1} \Sigma_t^G s_t^G = -\kappa_G \frac{G_t}{Q_t^1 B_t} \sum_k Q_t^k \Gamma^k \begin{bmatrix} \frac{\chi^2 \kappa_1}{\sigma_1^2} \\ \frac{\chi^2 \kappa_2}{\sigma_2^2} \\ \vdots \end{bmatrix}$$

The same steps apply to the portfolio that hedges the tax revenue risk. So

$$\Sigma_t^{-1} \Sigma_t^\Theta s_t^\Theta = \kappa_\Theta \kappa_j \frac{T_t}{Q_t^1 B_t} \sum_k Q_t^k \Gamma^k \begin{bmatrix} \frac{\chi^2 \kappa_1}{\sigma_1^2} \\ \frac{\chi^2 \kappa_2}{\sigma_2^2} \\ \vdots \end{bmatrix}.$$

The final portfolio we derive is the one that hedges the interest rate risk. The first step to get $\Sigma_t^Q s_t^Q$

$$\begin{aligned}\frac{\Sigma_t^Q s_t^Q[j]}{Q_t^1} &= \sum_{k=1}^{\infty} (\kappa_j \kappa_k \sigma_f^2 + \iota_{\{j=k\}} \sigma_j^2) s_t^Q[k] \\ &= \sum_{k=1}^{\infty} \iota_{\{j=k\}} \sigma_j^2 s_t^Q[k] + \kappa_j \sigma_f^2 \sum_{k=1}^{\infty} \kappa_k s_t^Q[k] \\ &= \sigma_j^2 s_t^Q[j] + \kappa_j \sigma_f^2 \sum_{k=1}^{\infty} \kappa_k s_t^Q[k]\end{aligned}$$

and then we multiply with Σ_t^{-1} to get

$$\left(\Sigma^{-1} \Sigma^Q s_T^Q\right)[j] = \frac{1}{\sum_{k=1}^{\infty} Q_t^k \Gamma^k} \left(Q_t^{j+1} \Gamma^{j+1} + \chi^2 \kappa_j \sigma_j^{-2} \left[\sum_{k=1}^{\infty} \kappa_k Q_t^{k+1} \Gamma^{k+1} - \sum_{k \in \mathcal{G}} \kappa_k Q_t^{k+1} \Gamma^{k+1} \right] \right).$$

So finally we get

$$\begin{aligned}\omega_t^X[j] &= \Sigma_t^{-1} \Sigma_t^{\Theta} s_t^{\Theta}[j] + \Sigma_t^{-1} \Sigma_t^G s_t^G[j] = \left(\frac{1}{1 - \hat{\beta}_t} \right) \left(\frac{\kappa_{\Theta} T_t - \kappa_G G_t}{Q_t^1 B_t} \right) \left(\frac{\kappa_j}{\sigma_j^2} \chi^2 \right), \\ \omega_t^Q[j] &= \left(\Sigma^{-1} \Sigma^Q s_T^Q \right)[j] = (1 - \hat{\beta}_t) \left[\left(Q_t^{j+1} \Gamma^{j+1} \right) + \left(\sum_{\ell \notin \mathcal{G}} Q_t^{\ell+1} \Gamma^{\ell+1} \kappa_{\ell} \right) \left(\frac{\kappa_j}{\sigma_j^2} \chi^2 \right) \right].\end{aligned}$$

C.4 Estimations and extrapolations

We estimate model our factor model (39) using OLS. In the main text (Table 2), we report the estimates for the baseline specification in which we restricted $\rho_G = \rho_{\Theta} = 1$ and $\rho_f = 0$. This estimation procedure produces estimates of $(\alpha_j, \rho_j, \kappa_j, \sigma_j^2)$ for eleven j , with the highest being $j = 40$. For constructing our target portfolios, we need to extrapolate (ρ_j, κ_j) for all $j > 1$. In the baseline extrapolation, we estimate δ_j and σ_j^2 by fitting the closest exponential function: $f(j) = e^0 - e^0 \exp(-e^1 \times j)$ for $f(j) \in \{\delta_j, \sigma_j^2\}$. We fit the parameters e^0 and e^1 to minimize sum of squares between fitted and actual values of δ_j and σ_j^2 . The fit is reported in Figure 3

Fit of covariances In the text we mention that a test for the factor model is how well it captures contemporaneous covariances. To implement this test we compute $cov(r^j, G)$ and $cov(r^j, \Theta)$ using the estimated factor model and plot it against the ones computed using the raw data for the 11 portfolios that we use. In Figure 4, we see the fit is good.

Components of the target portfolio In Figure 5, we breakup the target portfolio into the portfolio that hedges government spending ω_t^G , tax revenue risk ω_t^{Θ} , and interest rate risk ω_t^Q .

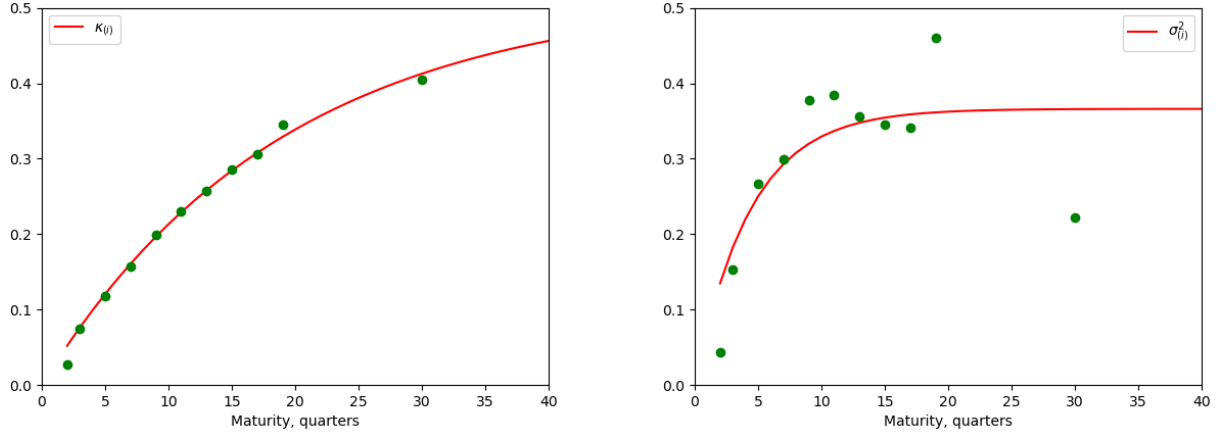


Figure 2: Fit for extrapolation of the factor model estimates of (κ_j, σ_j^2) using $f(j) = e^0 - e^0 \exp(-e^1 \times j)$ for factor model (39). The dotted points are the point estimates and the bold line is the interpolation.

Figure 3

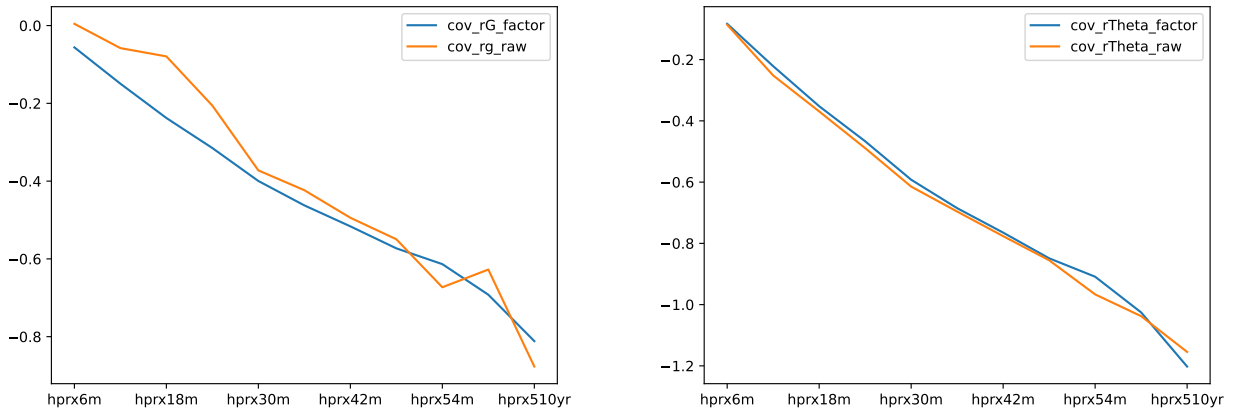


Figure 4: Fit for contemporaneous covariances. The blue lines is computed using the estimates of the factor model and the orange line is computed using the data for the sample period 1952-2017.

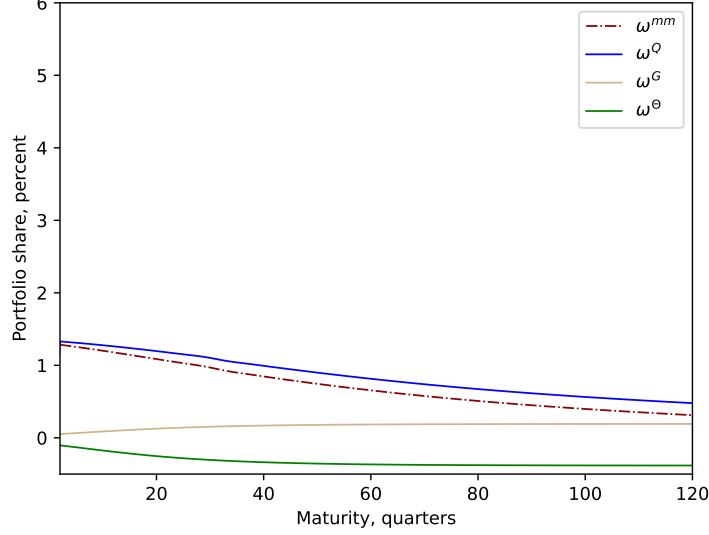


Figure 5: Components of the target portfolio.

C.5 Robustness

Factor-mimicking portfolios In the main text, our factor model was based on principal components extracted from returns and macroeconomic data. There is a tradition in the finance literature to use factor-mimicking portfolios. These portfolios are constructed by projecting macro variables on a set of returns and using the projection coefficients to construct portfolios. Practitioners then replace the poorly-measured macro variables with returns on those constructed portfolios. In our context extracting principal components from returns and projections on returns is the same as extracting them from only return data. In Figure 7, we report results using the common factor using only bond return data.

Multifactor We extend the factor model to have multiple factors. Let z_t be a stacked vector that consists of excess returns $\{r_t^j\}_j$ for the 11 portfolios of different maturities j , a measure of $\ln \Theta_t$ and expenditures $\ln G_t$. We use z_t^k to denote the k^{th} element of this vector. We posit the following stochastic process

$$\begin{aligned}
 z_t^k &= \alpha_k + \rho_k z_{t-1}^k + \sum_m \kappa_{m,k} f_t^m + \varepsilon_t^k \text{ for all } k, \\
 f_t^n &= \alpha_f^n + \sum_m \kappa_{m,f} f_t^m + \varepsilon_t^{f,n} \text{ for all } j = 1 \dots n,
 \end{aligned} \tag{20}$$

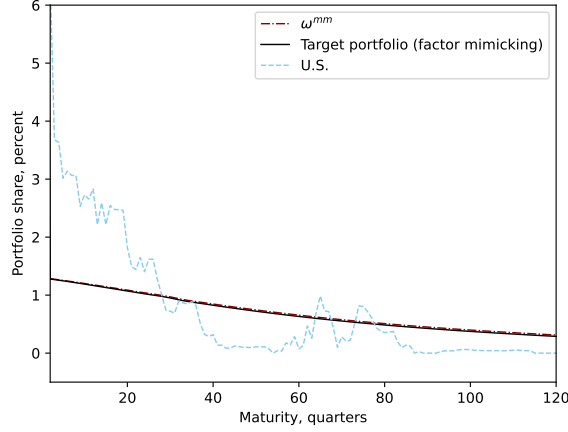


Figure 6: Target portfolio using the common factor extracted from bond excess returns.

Figure 7

where $\{f_t^n\}_n$ are a set of factors and $\{\varepsilon_t^k, \varepsilon_t^{f,n}\}_{k,t}$ are residuals. We use the subscripts $k \in \{\Theta, G\}$ to denote the variables $\ln \Theta_t$, $\ln G_t$, and $k = j$ to denote returns on bonds of maturity j . In this section, we report the estimates of the factor model and other details skipped in the main text.

With multiple factors the covariance of returns with each other is given by $\Sigma_t = \Delta + \kappa \Delta_f \kappa^\top$ where $\Delta = \text{diag}\{\sigma_j^2\}$, $\kappa = [\kappa_1 \ \kappa_2 \ \dots]$ is the matrix of factor loadings on returns and inverse Σ_t^{-1} can be obtained using Woodbury matrix identity. As before, we use Lemma 1 to compute Σ_t^Q using Σ_t . The covariances with spending and tax revenue risk are given by $\Sigma_t^G[j] = \sum_n \kappa_{n,j} \kappa_{n,G} \sigma_{f^n}^2$ and $\Sigma_t^\Theta[j] = \sum_n \kappa_{n,j} \kappa_{n,\Theta} \sigma_{f^n}^2$. The weights $\{s_t^Q, s_t^G, s_t^\Theta\}_t$ are unchanged.

In Figure 8, we plot the first two principal components extracted from excess returns, the risk-free rate, GDP (detrended), deficits/GDP. We see that the second factor is less volatile relative to the first factor. We next report the factor loadings for the two factor model in Table 1.

Fit of covariances Figure 9 plots the counterpart of Figure 4 for the multifactor model. We see that the fit is similar and slightly better than the single factor model.

Extrapolation of loadings In Figure 10, we plot the fit for the factor loadings for the multifactor model. We use the same form for the first factor as the baseline in the text. The second factor is non-monotonic and has a hump shape for intermediate maturities. To capture

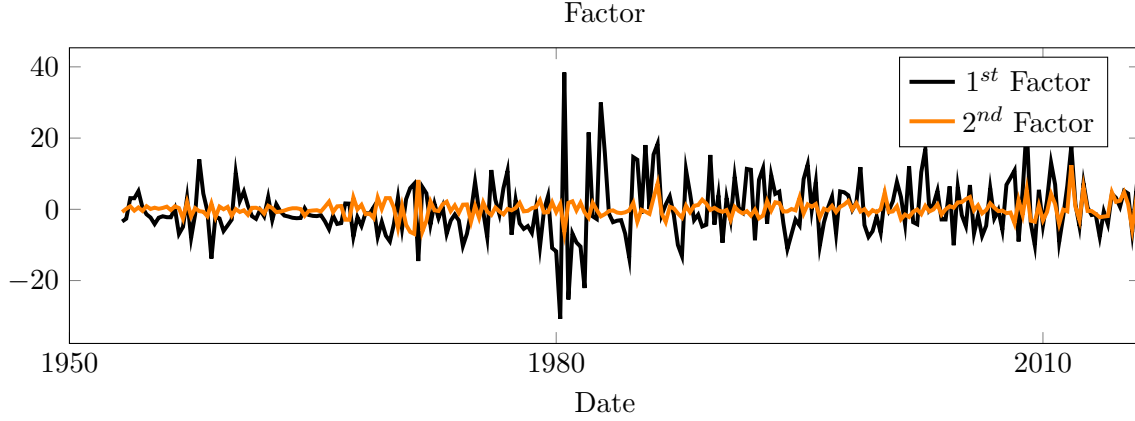


Figure 8: Time series for the first two principal components

Table 1: FACTOR LOADINGS (MULTIFACTOR)

	Excess returns r_t^j for various maturities j												
	6m	12m	18m	24m	30m	36m	42m	48m	54m	60m	120m	$\ln G_t$	$\ln \Theta_t$
$\kappa_{1,k}$	0.028	0.075	0.119	0.157	0.200	0.231	0.257	0.286	0.306	0.345	0.404	-0.032	-0.047
s.e	0.001	0.002	0.002	0.002	0.002	0.002	0.002	0.003	0.003	0.004	0.004	0.016	0.008
$\kappa_{2,k}$	-0.053	-0.126	-0.183	-0.208	-0.244	-0.248	-0.248	-0.218	-0.198	-0.184	-0.066	-0.031	0.024
s.e	0.005	0.008	0.009	0.008	0.008	0.008	0.006	0.009	0.011	0.015	0.013	0.058	0.030
σ_k^2	0.030	0.074	0.098	0.083	0.082	0.077	0.048	0.107	0.145	0.292	0.200	4.227	1.143
s.e	0.003	0.007	0.009	0.007	0.007	0.007	0.004	0.009	0.013	0.026	0.018	0.374	0.101
R^2	0.536	0.698	0.771	0.840	0.870	0.898	0.922	0.938	0.946	0.943	0.979	0.016	0.111

Notes: This table records the OLS factor loadings for the two factor version of the model (39). Standards errors are in parenthesis. The row titled “R2” are values of R-squared for each equation in the system (39). The sample for excess returns and primary surpluses normalized by outputs is 1952-2017, and the sample for the one-period liquidity premium is 1984-2017. The time period is a quarter.

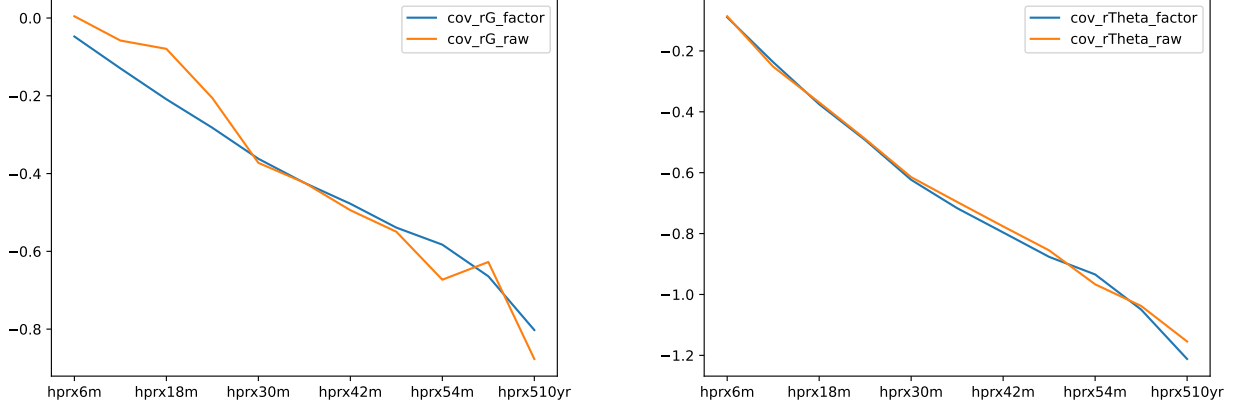


Figure 9: Fit for contemporaneous covariances. The blue lines is computed using the estimates of the multi-factor version of the factor model and the orange line is computed using the data for the sample period 1952-2017.

that hump, we use the functional form $e^0 + e^1 \exp(-(j - e^2)^2/e^3)$. We see in Figure 10 that the interpolated lines fit well the point estimates.

Limiting Portfolio with One Factor In the main text, we compared the limiting portfolio as we send the estimated $\sigma_j^2 \rightarrow 0$ for each j and keep the factor loadings $\{\kappa_{n,j}\}_{n,j}$ for the multifactor setting. In Figure (11), we plot the corresponding figure for the baseline target portfolio with one factor.

AR(1) factor structure We consider the general estimation of (39) without any a-priori restrictions on parameters. Table 2 presents estimation results. We see from the table that we cannot reject $\rho_G = \rho_\Theta = 1$ and $\rho_f = 0$.

Transitory Shock We assume that the spending is given by $G_t = G_t^p G_t^{tr}$ with G_t^p following the same structure as the baseline (39) and $G_t^{tr} = \rho_{tr} G_{t-1}^{tr}$. We then simulate the target portfolio for alternative values of the transitory $\{G_t^{tr}, \rho_{tr}\}$.

Such a shock affects the path of spending, the optimal tax rate, and through the optimal tax rate the expected path of tax revenues. The tax revenue risk weights are unchanged but the weights on spending risk $s_t^G[k] = \frac{-Q_t^k(\Gamma)^k G_t (G_t^{tr})^{1-\rho_{tr}^k}}{Q_t^1 B_t}$ and the weights on interest rate risk $s_t^Q[k] = \frac{Q_t^{k+1}(\Gamma)^{k+1} (T_t^{tax} - G_t (G_t^{tr})^{1-\rho_{tr}^{k+1}})}{Q_t^1 \sum Q_t^{k+1}(\Gamma)^{k+1} (T_t^{tax} - G_t (G_t^{tr})^{1-\rho_{tr}^{k+1}})}$ where T_t^{tax} are the tax revenues computed at the

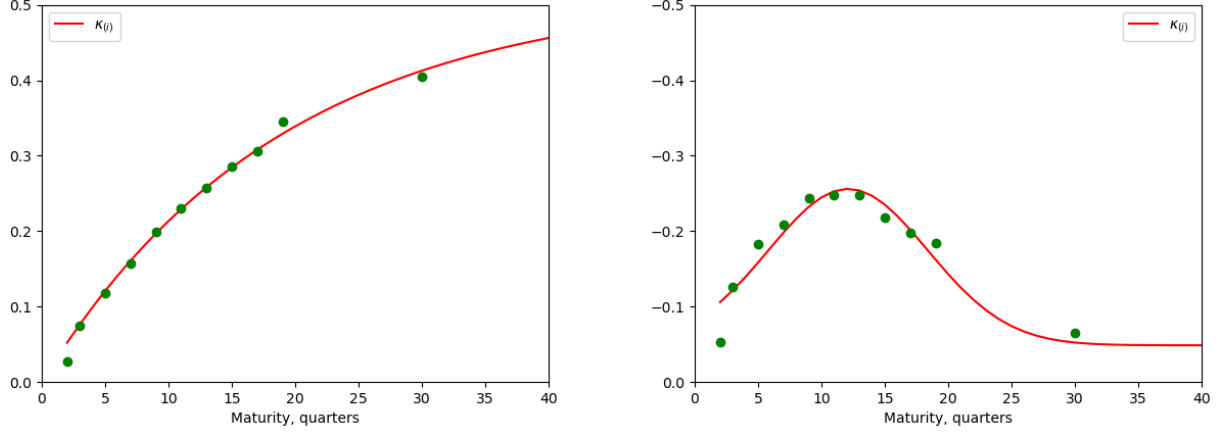


Figure 10: Fit for extrapolation of multifactor model estimates of $(\kappa_{1,j}, \kappa_{2,j})$. For the first factor, we use the functional form $f(j) = e^0 - e^0 \exp(-e^1 \times j)$ and for the second factor we use the functional form $(e^0 + e^1 \exp(-(j - e^2)^2/e^3))$. The dotted points are the point estimates and the bold line is the interpolation.

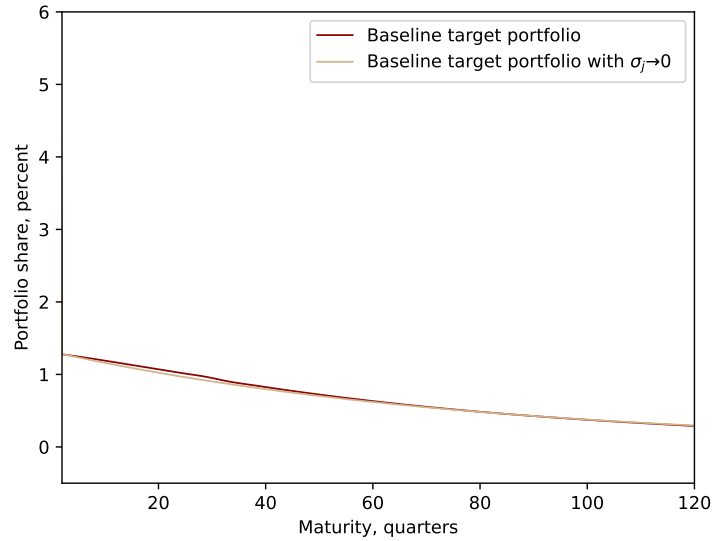


Figure 11: Comparison of the baseline target portfolio to the limiting portfolio in which we set $\sigma_j^2 \rightarrow 0$ for each j .

Table 2: FACTOR MODEL ESTIMATION RESULTS (AR(1) FACTOR STRUCTURE)

	Excess returns r_t^j for various maturities j													
	6m	12m	18m	24m	30m	36m	42m	48m	54m	60m	120m	$\ln G_t$	$\ln \Theta_t$	f_t
α_k	0.086 (0.014)	0.155 (0.025)	0.220 (0.033)	0.245 (0.035)	0.284 (0.039)	0.315 (0.039)	0.346 (0.038)	0.344 (0.037)	0.372 (0.037)	0.304 (0.043)	0.444 (0.030)	-0.177 (0.016)	-0.319 (0.008)	0.026 (0.502)
ρ_k	-0.107 (0.043)	-0.057 (0.035)	-0.041 (0.030)	-0.043 (0.025)	-0.042 (0.023)	-0.025 (0.020)	-0.022 (0.018)	-0.008 (0.016)	-0.022 (0.015)	-0.027 (0.015)	0.003 (0.009)	1.001 (0.008)	1.009 (0.004)	-0.035 (0.063)
κ_k	0.028 (0.002)	0.074 (0.003)	0.118 (0.004)	0.157 (0.004)	0.199 (0.005)	0.230 (0.005)	0.257 (0.005)	0.285 (0.005)	0.306 (0.005)	0.345 (0.005)	0.404 (0.004)	-0.032 (0.016)	-0.048 (0.008)	0.000 (nan)
σ_k^2	0.044 (0.004)	0.154 (0.014)	0.267 (0.024)	0.300 (0.027)	0.378 (0.034)	0.384 (0.034)	0.356 (0.031)	0.345 (0.031)	0.341 (0.030)	0.460 (0.041)	0.222 (0.020)	4.231 (0.375)	1.125 (0.100)	63.676 (5.65)
R2	0.536	0.698	0.771	0.840	0.870	0.898	0.922	0.938	0.946	0.943	0.979	0.985	0.996	0.001

Notes: This table records the OLS estimates of the factor model (39) without imposing $\rho_f = 0, \rho_\Theta = \rho_G = 1$. Standards errors are in parenthesis. The sample for excess returns and primary surpluses normalized by outputs is 1952-2017. The time period is a quarter.

optimal tax rate that is constant and balances the inter temporal budget at the zeroth order given the path of spending.

In left panel Figure 12, we show the target portfolio setting for a 10% and 20% increase in the share of spending to GDP with $\rho_{tr} = 0.75$ so that the increase lasts for about 5 years. In right panel, we plot the target portfolios for the same values of G_t^{tr} but a higher value of $\rho_{tr} = .95$.

Heteroskedastic shocks In the main text, we assumed that the shocks ε_t were homoskedastic, that is, we imposed that $\{\sigma_k\}$ for $k \in \{j, Y, G, A, f\}$ are constant through time. We relax that assumption and augment the baseline factor model (39) with the following univariate GARH processes $\{\sigma_k\}$

$$\sigma_{k,t}^2 = \bar{\sigma}_k^2 + \sum_{j=1}^p \rho_{kp}^{GARCH} \varepsilon_{zt-p}^2 + \sum_{j=1}^q \varrho_{kq}^{GARCH} \sigma_{\varepsilon z, t-q}^2$$

and impose that all ε are standard Gaussian and independent of each other. We now estimate the system using maximum likelihood and assuming $p = 2$ and $q = 1$.

The consequence of heteroskedastic shocks is that structure of the expressions for Σ_T and Σ_T^{-1} as well as Σ_T^k for $k \in \{X, A, Q\}$ remains the same but they have time-varying parameters $\sigma_{f,t}$ and $\sigma_{j,t}$ for each return maturities j .² We use the same extrapolation scheme as the baseline to obtain (σ_j, κ_j) for other maturities. And finally, as an implication, the optimal

²The time-variation in $\{\sigma_G^2, \sigma_Y^2, \sigma_A^2\}$ drops out because the covariances of hedging terms are driven by the common component captured in the factor $\{\sigma_{f,t}^2\}$.

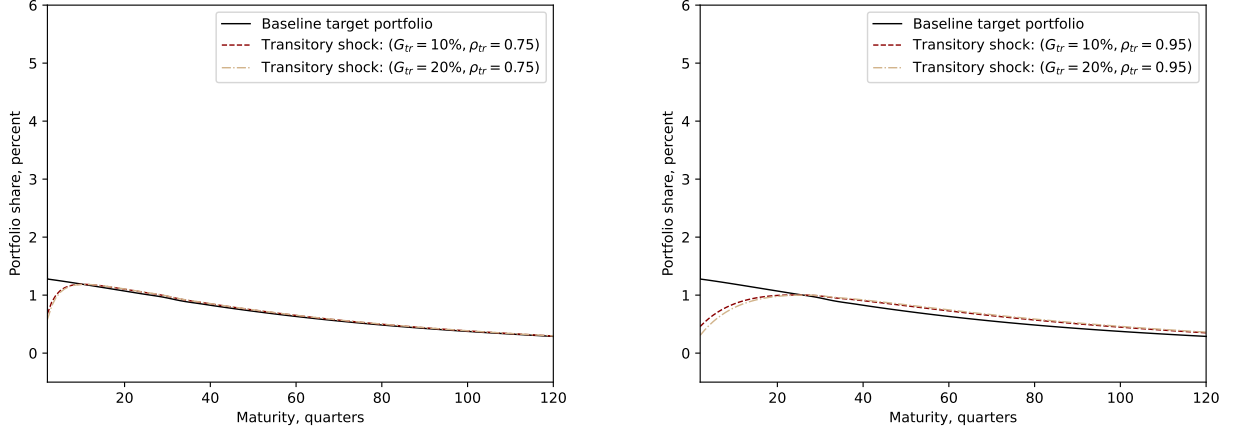


Figure 12: Comparison of optimal portfolios with a transitory increase in spending. In left panel we show the target portfolio for $\ln G_t^{tr} \in \{10\%, 20\%\}$ with $\rho_{tr} = 0.75$ and in right panel, we plot the target portfolios for $\ln G_t^{tr} \in \{10\%, 20\%\}$ with $\rho_{tr} = 0.95$.

target portfolio and its components also inherit that time-variation.³

In Figure 13, we plot the time-series for elements in $\{\sigma_{j,t}\}$ and $\sigma_{f,t}$. The volatilities for returns (including the factor) and macro aggregates are high in the early 80s and the great recession of 2008-2010 and quite stable in the intervening periods.

Keeping everything else the same, periods when the factor is more volatile increases the covariance of returns with each other as well as the covariance of returns with surpluses and liquidity risk. Thus, a priori the effect on the optimal portfolio is ambiguous. To gauge how much the portfolio moves overtime, we start by plotting in Figure 14), the 90-10 interval by maturity, that is, for each maturity we construct the 90th and 10th percentile across dates. We see that for lower maturities the portfolio shares varies a few basis points and the fluctuations are much smaller for larger maturities.

Alternative, time aggregation, calculation of returns We also experimented with alternative ways to calculate returns with different time frequencies. In the baseline, we used quarterly measures of returns, surpluses and taxes to ensure the largest sample such that we could measure asset prices and macro data in a consistent way. To verify if our results are driven by our choice of the frequency, we use returns and other macro variables at biannual frequencies. The shortest maturity available is now of 6 months, which we take as our measure of the one-period government bond R_t^{rf} . As before, we construct the biannual holding period

³In principle, the fiscal risk and liquidity risk portfolio could vary because quasi-weights π_T^X and π_T^A or $\vec{\beta}$ vary with time. To focus on the impact of heteroskedastic shocks, we keep them constant and equal to the values that we used in the main text and only allow the target portfolio to vary due to time-varying covariances.

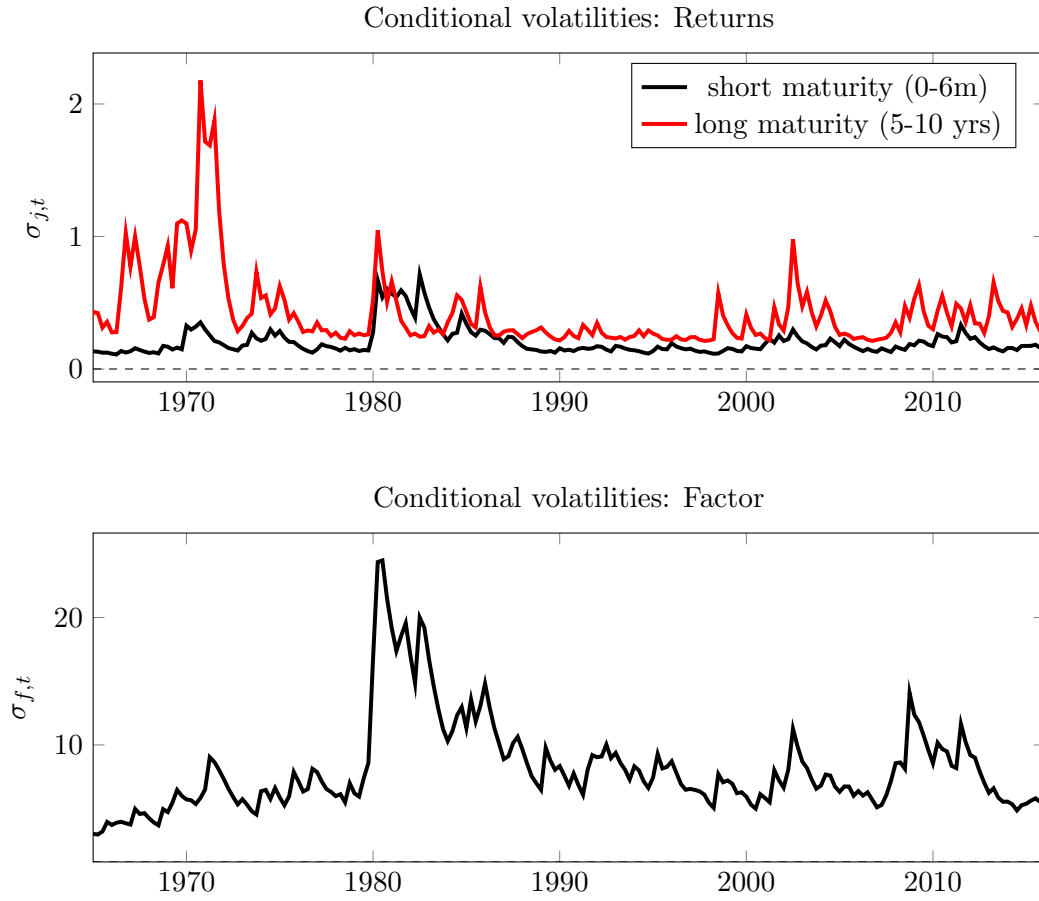


Figure 13: Conditional volatilities of returns, factor, using the estimated GARCH model

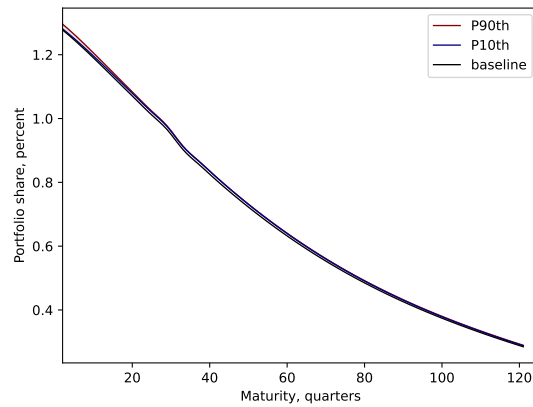


Figure 14: 90-10 interval of portfolio shares (maturities from 2 quarters to 120 quarters) with heteroskedastic shocks.

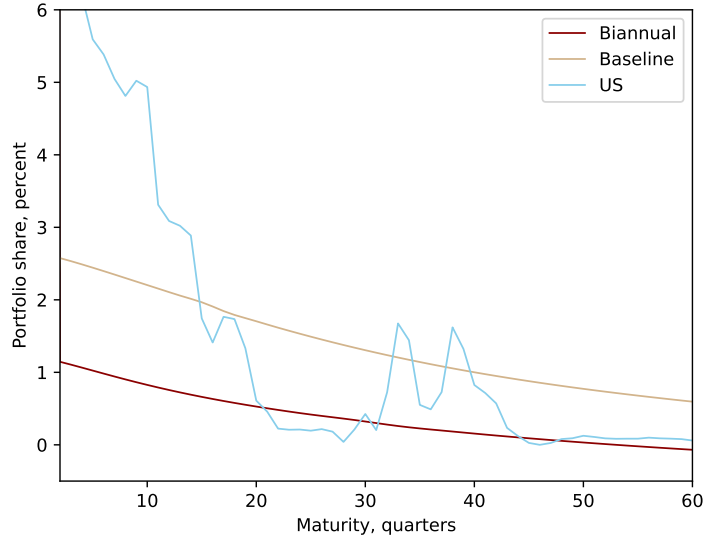


Figure 15: Bi-annual frequency

return by summing monthly returns for each portfolio which are separated by 6 month intervals. For other macro variables, we aggregate two consecutive quarters to obtain the biannual series. Using this data, we apply the same procedure as the baseline (extracting the factor, estimating the factor model, constructing the conditional covariances) and obtain the optimal portfolio. We show the implied unrestricted target portfolios in Figure 15. In order to compare it to our baseline results which have portfolios by quarterly bins, we aggregate the baseline portfolio weights to biannual weights using $\omega_{biannual}[i] \equiv \omega[2i - 1] + \omega[2i]$, where i indexes the 6 month intervals and the right hand side is the baseline target portfolio. We find that that the two biannual portfolios are similar.

C.6 Additional details for Section 5.3

In this appendix, we document how we estimate Λ_t using estimates from Greenwood and Vayanos (2014).

Greenwood and Vayanos report estimates of semi-elasticities $\left\{ Y_t \frac{\partial \ln \text{yields}_t^k}{\partial \text{Bond Supply}_t} \right\}_k$ by bond maturity for a subset of maturities using maturity-weighted debt $\sum_k k \tilde{B}_t^k$ as a measure of bond supply. We assume that weights $\tilde{\omega}_t^k = \frac{\tilde{B}_t^k}{\sum_k \tilde{B}_t^k}$ remain constant when $\sum_k k \tilde{B}_t^k$ changes. Under this assumption, a unit change in $\sum_k k \tilde{B}_t^k$ leads to a $\frac{1}{\tilde{\omega}_t^k}$ variation in supply of bond of maturity k . Using the definition of Λ_t from the text, and converting the estimate of yields to prices, we get

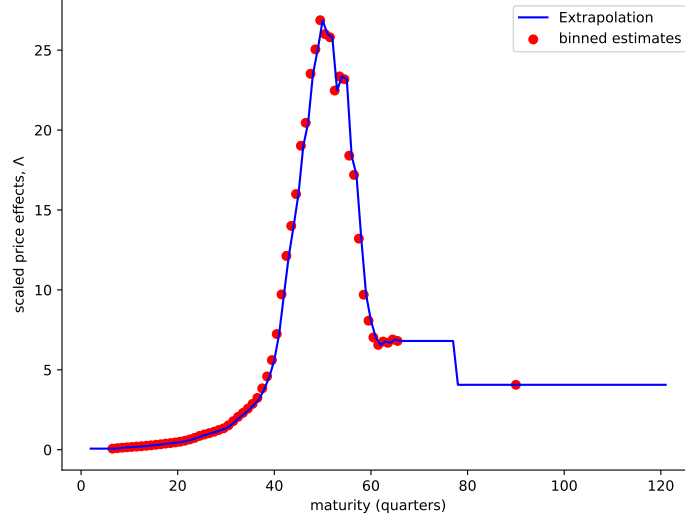


Figure 16: Fit for price effect semi-elasticities. We use 5 year bins with the last bin 15+ years to measure Λ . The red dots indicated estimates of the right hand side of equation 21 for the bins. The blue solid line is interpolated values that we use to measure all elements of Λ_t .

$$D_t \Lambda_t[j, k] = \iota_{\{k=j\}} \left(\frac{\sum_{k=1}^{\infty} Q_t^k}{(Q_t^1)^2} \frac{\xi^2(\tau_t)}{-\xi'(\tau_t)} \frac{1}{Q_t^k} \right) \left(\frac{k \times \left\{ Y_t \frac{\partial \ln \text{yields}_t^k}{\partial \text{Bond Supply}_t} \right\}_k}{\omega_t^{US,k}} \right) \left(\frac{\sum k \tilde{B}_t^{US,k}}{B_t} \right) \quad (21)$$

where all the terms on the right-hand side are measurable.⁴

The first term in (21) is obtained directly from estimates of the price curve $\{Q_t^k\}$ and the tax rate τ_t and we set them to the same values we use in Section 5. To get the second term in (21), we bin maturities in 5 year bins with the last bin for maturities greater than 15 years.

The red dots in Figure 16 are the implied values of $\left\{ \left(\frac{k \times \left\{ Y_t \frac{\partial \ln \text{yields}_t^k}{\partial \text{Bond Supply}_t} \right\}_k}{\omega_t^{US,k}} \right) \left(\frac{\sum k \tilde{B}_t^{US,k}}{B_t} \right) \right\}_k$. For intermediate maturities, we interpolate using the nearest value. The bold line in Figure 16 is the interpolated curve that we use in our implementation.

We then set Λ_t constant across dates at the estimated values and use equation (18) to

⁴Greenwood and Vayanos report elasticities $\left\{ Y_t \frac{\partial \ln \text{yields}_t^k}{\partial \text{Bond Supply}_t} \right\}_k$ for a subset of maturities. For an arbitrary maturity n , we assume that the elasticity satisfies $a_0 - a_0 \exp \{-a_1 n^2\}$ where parameters a_0 capture the long run level and parameter a_1 captures the speed of convergence. We fit (a_0, a_1) to match the long maturity elasticity of 0.003 and the short maturity (1 years) elasticity of 0.001 from Table 3 in Greenwood and Vayanos (2014). Since the Greenwood and Vayanos estimates are relatively flat over the maturities, the exact interpolation scheme is not critical of the results. We set $\left(\frac{\sum k \tilde{B}_t^k}{B_t} \right)$ to 7.3 using the values reported in Panel A of Table 1 in Greenwood and Vayanos (2014).

compute the steady state portfolio.

C.7 Additional details for Section 5.4

We can get a feel for how trading frictions affect the optimal portfolio by studying a special case. We assume stationarity and set $Q_t^k = \beta^k$, and further specialize to a simpler market structure in which the government trades only a risk-free security and a growth-adjusted consol. Let $\vec{\beta}$ be a geometrically declining series $\beta^k, \beta^{k-1}, \dots$ and excess return on the consol be denoted by r_t^∞ . Finally, we impose that the stochastic discount factor of the non-traders is scaled version of the stochastic discount factor of the traders: $\ln(M_{\mathbb{N},t+k}) = (1+\psi) \ln(M_{\mathbb{T},t+k})$. This introduces a new parameter, ψ , that captures the severity of trading frictions as $\psi > 0$ implies that the SDF of the non-traders is more volatile of those of the traders.

Under this last assumption the covariance of the relative stochastic discount factors simplifies to

$$\text{cov}_t \left(\ln(M_{\mathbb{T},t+k}) - \ln(M_{\mathbb{N},t+k}), r_{t+1}^j \right) = -\psi \text{cov}_t \left(\ln(M_{\mathbb{T},t+k}), r_{t+1}^j \right).$$

As the traders trade the consol, we can use the traders' Euler equation, equation (10), to substitute out for this covariance and obtain

$$-\text{cov}_t \left(\ln(M_{\mathbb{T},t+k}), r_{t+1}^j \right) \simeq \mathbb{E}_t r_{t+1}^j - \text{cov}_t \left(\ln Q_{t+1,t-1}, r_{t+1}^j \right).$$

Under our stationarity assumptions we have $\mu_{\mathbb{N},t+k} = \mu_{\mathbb{N},t}$ and can therefore express Σ_t^M as the sum of three terms

$$\Sigma_t^M = \mu_{\mathbb{N},t} \psi \left(\mathcal{R}_t - \bar{Q}_t^1 \Sigma_t^Q \right)$$

where $\mathcal{R}_t[j, k] = Q_t^1 \mathbb{E}_t r_{k+1}^j$.

The effect of non-traders on the optimal portfolio is given by $\Sigma_t^{-1} \Sigma_t^M s_t^{ineq}$. This simplifies under this market structure of a growth adjusted consol and a risk free bond as Σ_t is now a single number representing the conditional covariance of the growth adjusted consol. We can also make progress on the components of $\Sigma_t^M \vec{\beta}$, starting with $\mathcal{R}_t \vec{\beta} = \frac{\mathbb{E}_t r_{t+1}^j}{1-\beta}$. Next we note that

$$\begin{aligned} \Sigma_t^Q \vec{\beta} &= \sum_{k=1}^{\infty} \beta^k \text{cov}_t \left(\frac{1}{Q_t^1} \ln Q_{t+1,k}, r_{t+1}^\infty \right) \simeq \frac{\Gamma}{\beta} \sum_{k=1}^{\infty} \beta^k \mathbb{E}_t \partial_\sigma \ln Q_{t+1,k} \partial_\sigma r_{t+1}^\infty \\ &\simeq \frac{\Gamma}{\beta} \mathbb{E}_t \sum_{k=1}^{\infty} \Gamma^k \partial_\sigma Q_{t+1,k} \partial_\sigma r_{t+1}^\infty \\ &\simeq \frac{\Gamma}{\beta} \mathbb{E}_t \partial_\sigma Q_{t+1}^\infty \partial_\sigma r_{t+1}^\infty \\ &\simeq \frac{\Gamma}{1-\beta} \text{cov}_t(r_{t+1}^\infty, r_{t+1}^\infty) = \frac{\Gamma}{1-\beta} \Sigma_t. \end{aligned}$$

All put together we have that

$$\Sigma_t^{-1} \Sigma_t^M \vec{\beta} \approx \frac{\mu_{N,t} \psi}{1 - \beta} \left(\frac{\mathbb{E}_t r_{t+1}^\infty}{\text{var}_t(r_{t+1}^\infty)} - \beta \right).$$

So the presence of non-traders will lengthen the maturity as long as $\frac{\mathbb{E}_t r_{t+1}^\infty}{\text{var}_t(r_{t+1}^\infty)} > \beta$. We can construct estimates for both $\mathbb{E}_t r_{t+1}^\infty$ and $\text{var}_t(r_{t+1}^\infty)$ using the fact that the growth adjusted consol is the infinite sum of zero coupon bonds of all maturities weighted by β^j . To check the inequality, we use the estimates of the factor model and find that the $\frac{\mathbb{E}_t r_{t+1}^\infty}{\text{var}_t(r_{t+1}^\infty)}$ is indeed significantly larger than one and hence any reasonable estimate of β .

D Additional details for Section 6

To simulate the neoclassical model, we solve a complete markets Ramsey allocation as in Lucas and Stokey (1983) by posing the following maximization problem. Given some $t = 0$ state $s_0 \in \mathcal{S}$ and household savings $b_0(s^0)$, the Ramsey problem can be expressed as

$$\max_{c_t(s^t), y_t(s^t)} \mathbb{E}_0 \sum_{t=0}^{\infty} u \left(C_t, \frac{Y_t}{\theta_t} \right) \quad (22)$$

subject to

$$Y_t(s^t) = C_t(s^t) + G(s_t), \quad (23)$$

$$b_0(s^0) u_c(s^0) = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t) [u_c(s^t) C_t(s^t) + u_y(s^t) Y_t(s^t)], \quad (24)$$

where the *implementability constraints*, equation (24) is derived by taking the time-0 budget constraint and replacing after-tax wages as well as bond prices.

We assume that the state space $\mathcal{S}$ is discrete (described below) and non-linearly solve the optimal allocation using the first-order conditions of the Ramsey planning problem. The resulting optimal allocation is represented using two sets of vectors of dimension $2|\mathcal{S}|$, one set for consumption and labor choices at $t = 0$ and another set for all $s_t \in \mathcal{S}$ for $t \geq 1$. Using the Ramsey allocation $\{c_t, y_t\}$, we can back out other related objects such tax rates $\tau_t = 1 - \frac{\left(\frac{Y_t}{\theta_t}\right)^\gamma}{c_t^{1-\sigma}}$; primary surplus $X_t = \tau_t Y_t - G_t$; and zero-coupon bond prices $Q_t^n = \mathbb{E}_t \frac{C_{t+n}^{-\sigma}}{C_t^{-\sigma}}$.

We follow Buera and Nicolini (2004) and assume that the preferences of households are isoelastic $U \left(C_t, \frac{Y_t}{\theta_t} \right) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\left(\frac{Y_t}{\theta_t}\right)^{1+\gamma}}{1+\gamma}$ with parameters $\sigma = 2$ and $\gamma = 1$. The economy is closed, so the demand of assets from foreign investors is zero and there are liquidity services

provided by government bonds. The only source of uncertainty comes from the exogenous stochastic process of government expenditures G_t , which follows an AR(1) process

$$\ln G_t = \alpha_G + \rho_G \ln G_{t-1} + \sigma_G \epsilon_t$$

We set $(\alpha_G, \rho_G, \sigma_G)$ to obtain a mean G/Y of 15%, auto correlation of 0.95 and a standard deviation $\frac{1.2}{15}$ which are in line with the U.S. data that we use in Section 5.1. We discretize the $\ln G_t$ process with $|\mathcal{S}| = 20$ grid points. For our calculations, we set the level of initial debt B_0 so that the annualized initial level of government liabilities to GDP is 100%.

We use this parameterization to construct several versions of the optimal portfolio. First, for a given $s \in \mathcal{S}$, we apply Corollary to Theorem 1 in Angeletos (2002) and obtain the optimal portfolio $\omega_T^{CM,n}(s^T) = \omega^{CM,n}(s_T = s)$ for $n = 1, \dots, 20$ maturities that implements the complete markets allocation. We use the bond prices and present value of primary surpluses all of which can be backed out given the objects from the Ramsey allocation. In Figure 5, red color line, we plot $\{\omega^{CM,n}\}_{n=1}^{20}$ for $s_T = s_{10}$ which is the modal state.

Details for Figure 5 To obtain the target portfolio ω_t given some history s^t , we need to solve for a vector of portfolio shares that satisfies

$$\Sigma_t \omega_t = \left[\Sigma_t^Q s_t^Q + \Sigma_t^G s_t^G \right].$$

Before explaining how we get ω_t , we make two observations. First, given the properties of the Ramsey allocations, $\Sigma_t, \Sigma_t^Q s_t^Q, s_t^G, \Sigma_t^G$ only depend on state s_t , which we set to s_{10} and as before can be computed in closed form using the complete market allocation that we have already solved. Second, as mentioned in the main text the returns of different bonds are highly correlated in the neoclassical economy, which makes the matrix of returns Σ_t to be effectively non-invertible and there are a range of portfolios that satisfy inequality (42) for a given ϵ . To obtain the target portfolio that is plotted in Figure 5 blue color, we set $\epsilon = 1e - 8$ and pose the following minimization problem

$$\min_{\tilde{\omega}} \|\tilde{\omega} - \omega_t^{ABN}\|$$

such that

$$\left\| \Sigma_t \tilde{\omega} - \left[\Sigma_t^Q s_t^Q + \Sigma_t^G s_t^G \right] \right\| \leq \mathbf{1}^\top \epsilon.$$

where $\|\cdot\|$ we mean the sup norm. This formulation conveniently delivers an objective that is quadratic while the constraint set is linear and convex; and we use a standard methods (OSQP library) to solve the minimization problem.

E Closed Economy

In this appendix, we study a closed neoclassical version of our *benchmark economy*. Unlike the benchmark open economy specification in Section 2, a change in the governments portfolio will necessarily change the price of assets in economy; and, compared to the segmented markets version of the benchmark economy presented in Section 4.5, a change in the portfolio composition at date t will also affect the price of securities in all other periods.

In what follows, we show how to adjust our variational approach to incorporate such effects on prices. Our main result is to characterize the price effects and using that we show that the closed economy neoclassical setting implies price responses that are counterfactual relative to the evidence reviewed in Section 4.5. Besides the different structure on price effects, the rest of the analysis of a closed economy including the steps to obtain the expression for the optimal portfolio are similar to Section 3. In Section E.1, we formally describe the neoclassical closed economy environment that we study, then introduce the perturbation and analyze the welfare effects and optimality of the government. The proofs of the main results are in Section E.2.

E.1 Analysis

In addition to the assumptions of the benchmark economy we assume that:

1. Household preferences are time separable

$$V_t = u_t \left(C_t - \frac{(Y_t/\theta_t)^{1+1/\gamma}}{1 + 1/\gamma} \right) + \beta \mathbb{E}_t V_{t+1}.$$

2. Government expenditures $\{G_t\}$ are exogenous.
3. All assets are in zero net supply.⁵
4. The set of available securities can replicate a consol. We will let Q_t^∞ denote the price of the consol at date t .

Under these assumptions asset market clearing implies that

$$b_t^i = B_t^i$$

and

$$c_t + G_t = Y_t.$$

⁵That all assets are in zero net supply is for notational simplicity. Assuming positive net supply simply adds another term to the resource constraint equivalent to changing exogenous government expenditures.

Absence of trading frictions and non-pecuniary benefits of government securities the household optimality conditions imply

$$\mathbb{E}_t M_{t+1} R_{t+1}^i = M_t \text{ or } M_t Q_t^i = \mathbb{E}_t [M_{t+1} (d_{t+1}^i + Q_{t+1}^i)] \quad (25)$$

Perturbation Following Section 3, we use a perturbational approach to isolate the optimal public portfolio. The perturbations in this section are slightly different and adapted to get tractability in the closed economy environment.

We consider any competitive equilibrium and introduce a perturbation at a particular history s^t by assuming that the government purchases $\frac{\epsilon}{Q_t^i(s^t)}$ units of security j which is funded by selling $\frac{\epsilon}{Q_t^i(s^t)}$ of the risk free bond. This asset swap produces an additional $r_{t+1}^j(s^{t+1})\epsilon$ of excess returns at all histories s^{t+1} following s^t . We assume that the government uses those resources to purchase an additional $\frac{r_{t+1}^j(s^t)\epsilon}{1+Q_{t+1}^\infty(s^{t+1})}$ of the consol while keeping its holdings of all other assets constant. Due to its nature of swapping a longer security for a risk-free bond we will refer to this as a Quantitative Easing (or QE) perturbation and formally define it by

$$\partial_{QE} B_t^i(\tilde{s}^\ell) = \begin{cases} \frac{\epsilon}{Q_t^i(s^t)} & \text{if } i = rf \text{ and } \tilde{s}^\ell = s^t, \\ -\frac{\epsilon}{Q_t^j(s^t)} & \text{if } i = j \text{ and } \tilde{s}^\ell = s^t, \\ -\frac{1}{1+Q_{t+1}^\infty(s^t)} \left(r_{t+1}^j(s^{t+1}) \right) \epsilon & \text{if } i = \infty \text{ and } \tilde{s}^\ell \succ s^t, \ell > t, \\ 0 & \text{otherwise.} \end{cases}$$

The change in portfolio composition necessarily requires a change in taxes to balance the governments budget constraint,

$$G_t + \sum_i (Q_t^i + d_t^i) B_{t-1}^i = \tau_t Y_t + \sum_i Q_t^i B_t^i.$$

Differentiating with respect to ϵ in the direction of the QE perturbation yields the following response of tax revenues

$$-\partial_{QE} (\tau_\ell Y_\ell) = \frac{r_{t+1}^j(s^{t+1})}{1+Q_{t+1}^\infty(s^{t+1})} \left(I_{\{s^\ell \succ s^t\}} \right) + \sum \partial_\epsilon Q_\ell^i(s^\ell) \left(B_\ell^i(s^{\ell-1}) - B_{\ell-1}^i(s^\ell) \right) \quad (26)$$

where $I_{\{s^\ell \succ s^t\}}$ is an indicator returning 1 if history s^ℓ follows from s^t and zero otherwise. Intuitively the effect of the perturbation on tax revenues is a combination of two effects. The first, $\frac{r_{t+1}^j(s^{t+1})}{1+Q_{t+1}^\infty(s^{t+1})} \left(I_{\{s^\ell \succ s^t\}} \right)$, are the direct effects that are a result of the excess returns generated by the asset swap. The second, $\sum \partial_{QE} Q_\ell^i(s^\ell) \left(B_\ell^i(s^{\ell-1}) - B_{\ell-1}^i(s^\ell) \right)$, is the indirect effect that arises because the asset swap in period t changes prices not only in all future periods but also in all past periods starting from the initial date 0.

Assuming that the equilibrium manifold is sufficiently smooth, we can apply the envelope theorem to the household's maximization problem to obtain the welfare impact of this perturbation as $\epsilon \rightarrow 0$. The welfare effect of this perturbation comes from its effect on both tax rates and security prices and is given by

$$\begin{aligned}
\partial_{QE} V_0 &= \mathbb{E}_0 \sum_{\ell \geq 0} M_\ell \left(-\frac{\partial_{QE} (\tau_\ell Y_\ell)}{\xi_\ell} + \sum_{i \geq 0} \partial_{QE} Q_\ell^i (b_{\ell-1}^i - b_\ell^i) \right) \\
&= \mathbb{E}_0 \sum_{\ell \geq 0} M_\ell \left(-\frac{\partial_{QE} (\tau_\ell Y_\ell)}{\xi_\ell} + \sum_{i \geq 0} \partial_{QE} Q_\ell^i (B_{\ell-1}^i - B_\ell^i) \right) \\
&= \mathbb{E}_0 \left[\sum_{\ell \geq 0} M_\ell \sum_{i \geq 0} \partial_{QE} Q_\ell^i \left(\frac{\xi_\ell B_{\ell-1}^i - B_{\ell-1}^i}{\xi_\ell} - \frac{\xi_\ell B_\ell^i - B_\ell^i}{\xi_\ell} \right) + \sum_{\ell \geq T+1} \left(\frac{M_\ell}{\xi_\ell} \right) (I_{\{\bar{s}^\ell \succ s^T\}}) \left(\frac{r_{T+1}^j}{1 + Q_{T+1}^\infty} \right) \right] \\
&= \mathbb{E}_0 \left[\sum_{\ell \geq 0} M_\ell \left(\frac{\xi_\ell - 1}{\xi_\ell} \right) \sum_{i \geq 0} \partial_{QE} Q_\ell^i (B_{\ell-1}^i - B_\ell^i) + \sum_{\ell \geq t+1} \left(\frac{M_\ell}{\xi_\ell} \right) (I_{\{\bar{s}^\ell \succ s^t\}}) \left(\frac{r_{t+1}^j}{1 + Q_{t+1}^\infty} \right) \right] \\
&= \text{Pr}_0(s^t) M_t(s^t) \left[PE + \mathbb{E}_t \sum_{k \geq 1} \left(\frac{M_{t+k}}{M_t} \right) \left(\frac{r_{t+1}^j}{1 + Q_{t+1}^\infty} \right) \frac{1}{\xi_{t+k}} \right] \tag{27}
\end{aligned}$$

with

$$PE = \frac{1}{\text{Pr}_0(s^t) M_T(s^t)} \mathbb{E}_0 \left[\sum_{\ell \geq 0} M_\ell \left(\frac{\xi_\ell - 1}{\xi_\ell} \right) \sum_{i \geq 0} \partial_{QE} Q_\ell^i (B_{\ell-1}^i - B_\ell^i) \right].$$

The term $\mathbb{E}_t \sum_{k \geq 1} \left(\frac{M_{t+k}}{M_t} \right) \left(\frac{r_{t+1}^j}{1 + Q_{t+1}^\infty} \right) \frac{1}{\xi_{t+k}}$ parallels the effect of the same perturbation in the open economy benchmark model, and can be analyzed in a similar manner. Now, in addition to that term, we also have PE that captures the effect on asset prices for all histories starting from time 0 onward. In the next section we will show how our second order expansions can allow us express that term using covariances that can be measured in the data.

Characterizing the Price Effects The perturbation affects asset prices through its effect on the stochastic discount factor of the household. This can be seen by differentiating the household Euler equation (25) with respect to ϵ in the direction of the perturbation to get

$$(\partial_{QE} M_\ell) Q_\ell^i + M_\ell (\partial_{QE} Q_\ell^i) = \mathbb{E}_\ell [\partial_{QE} M_{\ell+1} (d_{\ell+1}^i + Q_{\ell+1}^i) + M_{\ell+1} (\partial_{QE} Q_{\ell+1}^i)].$$

As the perturbation affects the stochastic discount factor through changes in tax rates we define $\xi_{M,\ell} \equiv \frac{\partial \log M_\ell}{\partial (\tau_\ell Y_\ell)}$ as the semi-elasticity of $\log M_t$ with respect to the tax revenues which implies $\partial_{QE} M_\ell = M_\ell \xi_{M,\ell} \partial_{QE} (\tau_\ell Y_\ell)$. Under our assumptions, this semi-elasticity is given by

$$\xi_{M,\ell} = -\psi_\ell \times \frac{1}{Y_\ell - G_\ell - \theta_\ell v(Y_\ell)} \times \left(\frac{\xi_\ell - 1}{\xi_\ell} \right)$$

where $\psi_\ell \equiv \frac{-[c_\ell - v_\ell(Y_\ell)]U''(c_\ell - v_\ell(Y_\ell))}{U'(c_\ell - v_\ell(Y_\ell))}$ is the coefficient of relative risk aversion.

To get a better understanding of how these terms contribute the price effects in the closed economy we'll focus on a stationary version of the economy

Definition 1. An optimal competitive equilibrium is *stationary from time t* if there exists a constant R_t such that for all $\ell > t$ (i) $\mathbb{E}_t G_\ell \approx G_t$ (iii) $\mathbb{E}_t R_\ell^i \approx R_t$ for all i and (iv) $\mathbb{E}_t c_\ell \approx c_t$.

This definition of stationary differs from the stationarity of the main text in that we assume a growth rate of $\Gamma = 1$. All of our results extend to a positive growth rate assuming that the utility function is CRRA.⁶ Our first set of results concern the asset pricing implications of the QE perturbation. We will leave the proof of both propositions to the end of the section.

Proposition 1. *For a neoclassical model which is stationary from time t*

1. The QE perturbation keeps asset prices zero to the first-order

$$\partial_\sigma \partial_{QE} Q_\ell^i = 0 \quad \forall \quad i, \ell \geq 0$$

2. The QE perturbation only affects risk-premia at t

$$\mathbb{E}_\ell \partial_{\sigma\sigma} \partial_{QE} r_{\ell+1}^i = 0 \quad \forall \quad \ell \neq t$$

and at date ℓ

$$\mathbb{E}_\ell \partial_{\sigma\sigma} \partial_{QE} r_{\ell+1}^i = \frac{2\psi_t}{Y_t - G_t - \theta_t v(Y_t)} \times \left(\frac{1 - \xi_t}{\xi_t} \right) \left(\frac{1}{1 + Q_{t+1}^\infty} \right) \mathbb{E}_\ell \partial_\sigma r_{\ell+1}^j \partial_\sigma r_{\ell+1}^i > 0,$$

where ψ_t is coefficient of relative risk aversion.

This proposition states that the QE perturbation does not effect prices to zeroth or first order. This is inline with our modeling of price effects in Section 4.5 where we assume that the effect prices is at second order. Intuitively, to zeroth and first-order all assets have the same expected return so the QE perturbation only changes the risk profile of the household's stochastic discount factor which, in turn, will only effect prices to second order. Moreover, the proposition states that the effect on asset prices in the closed economy are counterfactual to what has been documented in the data. Estimates by Greenwood and Vayanos (2014) and others find that find that $\Lambda^{QE}[rf, j] \approx 0$ and $\Lambda^{QE}[i, j] > 0$ for $i > rf$ which implies that expected excess returns should decrease with the QE perturbation rather than increase:

$$\mathbb{E}_t \partial_{\sigma\sigma} \partial_{QE} r_{t+1}^i = -\frac{\bar{Q}_{t+1}^i}{\bar{Q}_t^i} \frac{\partial_{\sigma\sigma} \partial_{QE} Q_t^i}{\bar{Q}_t^i} < 0.$$

⁶The main difference is that we will require that the government smooth excess returns using a growth-adjusted consol rather than a pure consol.

When governments buy back long term debt by issuing short term debt, short term rates appear to be unchanged so expected excess returns are driven by the fall in the term premia as the increased demand drives up prices. In contrast, in the closed economy, the government returns the excess returns from the QE swap via taxes which results in making states of the world where excess returns are high (low) better (worse) for the household by lowering (raising) tax rates in those states. As a result, the value of the asset decreases which raises the risk-premia. As noted, this is inconsistent with the segmented market literature which finds that the excess returns on long maturity debt are lower after QE. Finally, we are able to use our expansions to characterize the price effects

Proposition 2. *For a neoclassical economy which is stationary from time 0, if all initial debt $\{B_{-1}^i\}_i$ was risk-free then $PE \simeq \left(\frac{\bar{\xi}}{\bar{\xi}-1}\right)^{-1} \Psi_t(s^t)$ where*

$$\begin{aligned} \Psi_t(s^t) = & \frac{-2\bar{B}\bar{\xi}_M(\bar{Q}^1 - 1)}{(1 - \bar{B}(\bar{Q}^1 - 1)\bar{\xi}_M)} \sum_{\ell=t+1}^{\infty} \left(\frac{(\bar{Q}_t^1)^{\ell-t}}{1 + \bar{Q}_{t+1}^{\infty}} \right) cov_t \left(\partial_{\sigma} \ln M_{\ell}, \partial_{\sigma} r_{t+1}^j \right) \\ & - \frac{2\bar{\xi}_M \bar{B}}{(1 - \bar{B}(\bar{Q}^1 - 1)\bar{\xi}_M)} \sum_{\ell=t+1}^{\infty} \left(\frac{(\bar{Q}_t^1)^{-t}}{1 + \bar{Q}_{t+1}^{\infty}} \right) cov_t \left(\partial_{\sigma} r_{t+1}^j, \partial_{\sigma} \ln Q_{\ell}^1 \right) \\ & - \frac{2\bar{\xi}_M}{(1 - \bar{B}(\bar{Q}^1 - 1)\bar{\xi}_M)} \sum_{j \geq 1} \frac{\bar{Q}_t^1}{1 + \bar{Q}_{t+1}^{\infty}(\ell^{t+1})} cov_t \left(\partial_{\sigma} r_{t+1}^j, \partial_{\sigma} r_{t+1}^j \right) \\ & - \frac{2\bar{B}}{(1 - \bar{B}(\bar{Q}^1 - 1)\bar{\xi}_M)} \sum_{\ell=t}^{\infty} \left(\frac{(\bar{Q}_{\ell}^1)^{\ell-t}}{1 + \bar{Q}_{t+1}^{\infty}} \right) cov_t \left(\partial_{\sigma} \xi_{M,\ell} - \partial_{\sigma} \xi_{M,\ell+1}, \partial_{\sigma} r_{t+1}^j \right) \end{aligned}$$

As we have noted without any assumptions price effects are given by

$$PE = \frac{1}{Pr_0(s^t) M_T(s^t)} \mathbb{E}_0 \left[\sum_{\ell \geq 0} M_{\ell} \left(\frac{\xi_{\ell} - 1}{\xi_{\ell}} \right) \sum_{i \geq 0} \partial_{QE} Q_{\ell}^i (B_{\ell-1}^i - B_{\ell}^i) \right]$$

where a swap of securities at a particular history can affect asset prices at all other histories—past and future—due to general equilibrium effects on the stochastic discount factor that now directly depends on the tax rates. Proposition 2 allows us to characterize these price effects with a closed form expression using entirely time t covariances that are measurable in the data.

E.2 Proofs for Propositions 1 and 2

E.2.1 Proof of Proposition 1

We begin by noting that at the zeroth order, we get $\bar{\xi}_{M,\ell}^j = -\frac{\bar{\psi}_{\ell}}{\bar{Y}_{\ell} - \bar{G}_{\ell} - \bar{\theta}_{\ell} v(\bar{Y}_{\ell})} \times \left(\frac{\bar{\xi}_{\ell} - 1}{\bar{\xi}_{\ell}} \right) = \bar{\xi}_{M,t}$, is independent of time and the details of the perturbation. We proceed by proving a series of

lemmas documenting the results of Proposition 1

Lemma 1. *Expected excess returns are zero to the zeroth and the first order*

Proof. The zeroth of (25) gives us

$$\bar{r}_{\ell+1}^i = 0$$

Take first order expansion to get

$$\mathbb{E}_\ell \partial_\sigma r_{\ell+1}^i \overline{M_{\ell+1}} + \mathbb{E}_\ell \overline{r_{\ell+1}^i} \partial_\sigma M_{\ell+1} = 0$$

and thus

$$\mathbb{E}_\ell \partial_\sigma r_{\ell+1}^i = 0.$$

□

Lemma 2. *To the first order, price effects are zero, that is, for all i, ℓ : $\partial_\sigma \partial_{QE} Q_\ell^i = 0$*

Proof. Start from the definition of Q_ℓ^i

$$Q_\ell^i(s^\ell) = \mathbb{E}_{s^\ell} \sum_{k \geq 1} \frac{M_{\ell+k}}{M_\ell} D_{\ell+k}^i.$$

$$\partial_\sigma \partial_{QE} Q_\ell^i = \mathbb{E}_\ell \sum_{k \geq 1} (\partial_{QE} \partial_\sigma \log M_{\ell+k} - \partial_{QE} \partial_\sigma \log M_\ell) \left(\frac{\overline{M_{\ell+k}}}{\overline{M_\ell}} \right) D_{\ell+k}^i.$$

A necessary and sufficient condition for price effects to be zero at the first order is that $k \geq 1$

$$\mathbb{E}_\ell (\partial_{QE} \partial_\sigma \log M_{\ell+k} - \partial_{QE} \partial_\sigma \log M_\ell) = 0 \quad (28)$$

Use the definition of $\xi_M(s^\ell)$ to get $\partial_{QE} \log M_\ell(s^\ell) = \partial_{QE}(\tau_\ell(s^\ell) Y_\ell(s^\ell)) \times \xi_M(s^\ell)$. To first order

$$\partial_\sigma \partial_{QE} \log M_\ell = \partial_\sigma \partial_{QE}(\tau_\ell Y_\ell) \times \bar{\xi}_{M,\ell}$$

Then (28) is equivalently expressed as

$$\mathbb{E}_\ell (\partial_\sigma \partial_{QE} \log M_{\ell+k} - \partial_\sigma \partial_{QE} \log M_\ell) = \bar{\xi}_{M,\ell} (\mathbb{E}_\ell \partial_\sigma \partial_{QE}(\tau_{\ell+k} Y_{\ell+k}) - \partial_\sigma \partial_{QE}(\tau_\ell Y_\ell))$$

We check condition (28) by guess and verify.

Suppose $\partial_\sigma \partial_{QE} Q_\ell^i = 0$ for $\ell \geq 0$, then for all $\ell \geq 0$ and from equations (26)

$$-\partial_\sigma \partial_{QE}(\tau_\ell Y_\ell) = \partial_\sigma \left(\frac{r_{t+1}^j(s^{t+1})}{1 + Q_{t+1}^\infty(s^{t+1})} \right) I_{\{s^\ell \succ s^{t+1}\}} = \frac{\partial_\sigma r_{t+1}^j(s^{t+1})}{1 + Q_{t+1}^\infty} I_{\{s^t \succ s^{t+1}\}}$$

When $\ell \geq t+1$

$$\mathbb{E}_\ell (\partial_\sigma \partial_{QE} \log M_{t+1+k} - \partial_\sigma \partial_{QE} \log M_{t+1}) = \bar{\xi}_{M,t+1} \left(\frac{\partial_\sigma r_{t+1}^j(s^{t+1})}{1 + \bar{Q}_{t+1}^\infty} - \frac{\partial_\sigma r_{t+1}^j(s^{t+1})}{1 + \bar{Q}_{t+1}^\infty} \right) I_{\{s^\ell \succ s^{t+1}\}} = 0$$

When $\ell \leq t$, we can use the fact that to the first order, expected excess returns are zero from Lemma (2) to establish that (28) holds. $\square$

Lemma 3. *In the closed economy the effect of the perturbation on expected excess returns is*

$$\mathbb{E}_\ell \partial_{\sigma\sigma} \partial_{QE} r_{\ell+1}^i = 0 \quad \forall \quad \ell \neq t$$

and at date t

$$\mathbb{E}_t \partial_{\sigma\sigma} \partial_{QE} r_{t+1}^i = \frac{2\psi_t}{Y_t - G_t - \theta_t v(Y_t)} \times \left(\frac{1 - \xi_t}{\xi_t} \right) \left(\frac{1}{1 + \bar{Q}_{t+1}^\infty} \right) \mathbb{E}_\ell \partial_\sigma r_{\ell+1}^j \partial_\sigma r_{\ell+1}^i > 0$$

Proof. The first order expansion $\partial_{QE} M_\ell$ after using Lemma 2 gives us

$$\partial_\sigma \partial_{QE} M_{\ell+1} = -\bar{\xi}_{M,\ell} \bar{M}_{\ell+1} \left\{ \partial_\sigma \left(\frac{r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \right) I_{\{s^\ell \succ s^{t+1}\}} \right\}$$

Use this along with the second order expansion of households optimality condition (25) to obtain

$$0 = \mathbb{E}_\ell \partial_\sigma r_{\ell+1}^i \left(-\bar{\xi}_{M,\ell} \bar{M}_{\ell+1} \left\{ \partial_\sigma \left(\frac{r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \right) I_{\{s^\ell \succ s^{t+1}\}} \right\} \right) + \mathbb{E}_\ell \partial_{\sigma\sigma} \partial_{QE} r_{\ell+1}^i \bar{M}_{\ell+1}$$

For $\ell < t$, $I_{\{s^\ell \succ s^{t+1}\}} = 0$ and thus $\mathbb{E}_\ell \partial_{\sigma\sigma} \partial_{QE} r_{\ell+1}^i = 0$.

For $s^\ell \succ s^{t+1}$, use Law of iterated expectations to get

$$0 = \mathbb{E}_{t+1+k} \partial_\sigma r_{t+1+k}^i \left(-\bar{\xi}_{M,t+1+k} \bar{M}_{t+1+k} \underbrace{\mathbb{E}_{t+1} \left\{ \partial_\sigma \left(\frac{r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \right) \right\}}_{=0} \right) + \mathbb{E}_{t+1+k} \partial_{\sigma\sigma} \partial_{QE} r_{t+k+2}^i \bar{M}_{t+2+k}$$

and use Lemma (2) to get $\mathbb{E}_\ell \partial_{\sigma\sigma} \partial_{QE} r_{\ell+1}^i = 0$ for $s^\ell \succ s^{t+1}$.

Finally for $\ell = t$

$$0 = \mathbb{E}_t \partial_\sigma r_{t+1}^i \left(-\bar{\xi}_{M,t} \bar{M}_{t+1} \left\{ \partial_\sigma \left(\frac{r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \right) \right\} \right) + \mathbb{E}_\ell \partial_{\sigma\sigma} \partial_{QE} r_{\ell+1}^i \bar{M}_{t+1}.$$

Substitute for $\bar{\xi}_{M,t}$ and simplify to get

$$\mathbb{E}_t \partial_{\sigma\sigma} \partial_{QE} r_{t+1}^i \simeq \frac{2\psi_t}{Y_t - G_t - \theta_t v(Y_t)} \times \left(\frac{1 - \xi_t}{\xi_t} \right) \left(\frac{1}{1 + \bar{Q}_{t+1}^\infty} \right) \mathbb{E}_\ell \partial_\sigma r_{t+1}^j \partial_\sigma r_{\ell+1}^i.$$

Since $\xi_t = 1 - \gamma \frac{\tau_t}{1 - \tau_t} < 1$, $Y_t - G_t - \theta_t v(Y_t) > 0$ from Inada conditions, and $\psi_t > 0$, we get that $\mathbb{E}_t \partial_{\sigma\sigma} \partial_{QE} r_{t+1}^j > 0$. $\square$

E.2.2 Proof of Proposition 2

The second order expansion of the price effects

$$\partial_{\sigma\sigma} (\text{Pr}_0(s^t) M_t(s^t) PE_{j,t,\epsilon}) = \mathbb{E}_0 \left[\sum_{\ell \geq 0} \left(\frac{\bar{\xi}_\ell - 1}{\bar{\xi}_\ell} \right) M_\ell \sum_{i \geq 0} \partial_{\sigma\sigma} \partial_{QE} Q_\ell^i (\bar{B}_{\ell-1}^i - \bar{B}_\ell^i) \right] \quad (29)$$

which equals

$$\left(\frac{\bar{\xi}_0 - 1}{\bar{\xi}_0} \right) \bar{M}_0 \sum_{i \geq 0} \partial_{\sigma\sigma} \partial_{QE} Q_0^i \bar{B}_{-1}^i + \left(\frac{\bar{\xi}_0 - 1}{\bar{\xi}_0} \right) \mathbb{E}_0 \left[\sum_{t \geq 0} \sum_{i \geq 0} \bar{B}_t^i (\bar{M}_{t+1} \partial_{\sigma\sigma} \partial_{QE} Q_{t+1}^i - \bar{M}_t \partial_{\sigma\sigma} \partial_{QE} Q_t^i) \right]. \quad (30)$$

Its easy to see that $\left(\frac{\bar{\xi}_0 - 1}{\bar{\xi}_0} \right) \bar{M}_0 \sum_{i \geq 0} \partial_{\sigma\sigma} \partial_{QE} Q_0^i \bar{B}_{-1}^i = \left(\frac{\bar{\xi}_0 - 1}{\bar{\xi}_0} \right) \bar{M}_0 \sum_{i \neq 1} \partial_{\sigma\sigma} \partial_{QE} Q_0^i \bar{B}_{-1}^i = 0$ under the assumption that initial debt was risk-free.

the household pricing equation implies

$$M_\ell Q_\ell^i = \mathbb{E}_\ell [M_{\ell+1} (Q_{\ell+1}^i + D_{\ell+1}^i)] \quad (31)$$

Differentiating by ∂_{QE} gives

$$(\partial_{QE} M_\ell) Q_\ell^i + M_\ell \partial_{QE} Q_\ell^i = \mathbb{E}_\ell [(\partial_{QE} M_{\ell+1}) (Q_{\ell+1}^i + D_{\ell+1}^i) + M_{\ell+1} \partial_{QE} Q_{\ell+1}^i]$$

Let's start by looking at $\ell < t$, We know that $\partial_\sigma \partial_{QE} M_{\ell+1} = 0$ so taking the second derivative with respect to σ yields

$$\mathbb{E}_\ell [\bar{M}_{\ell+1} \partial_{\sigma\sigma} \partial_{QE} Q_{\ell+1}^i - \bar{M}_\ell \partial_{\sigma\sigma} \partial_{QE} Q_\ell^i] = \bar{Q}_\ell^i \mathbb{E}_\ell [(\partial_{\sigma\sigma} \partial_{QE} M_\ell) - (\partial_{\sigma\sigma} \partial_{QE} M_{\ell+1}) \bar{R}_{\ell+1}^{rf}].$$

For $\ell > t$ and $s^\ell \succ s^t$ we have $\frac{\partial_{QE} M_\ell}{\bar{M}_\ell} = \xi_{M,\ell}^j \partial_{QE} (\tau_\ell Y_\ell)$ and hence $\partial_\sigma \partial_{QE} M_\ell = \bar{M}_\ell \bar{\xi}_{M,\ell} \frac{\partial_\sigma r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty}$.

The second order expansion of equation (31) is

$$2\partial_\sigma \partial_{QE} M_\ell \partial_\sigma Q_\ell^i + (\partial_{\sigma\sigma} \partial_{QE} M_\ell) \bar{Q}_\ell^i + \bar{M}_\ell \partial_{\sigma\sigma} \partial_{QE} Q_\ell^i = \mathbb{E}_\ell [2\partial_\sigma \partial_{QE} M_{\ell+1} \partial_\sigma (Q_{\ell+1}^i + D_{\ell+1}^i)] \\ + (\partial_{\sigma\sigma} \partial_{QE} M_{\ell+1}) (\bar{Q}_{\ell+1}^i + D_{\ell+1}^i) + \bar{M}_{\ell+1} \partial_{\sigma\sigma} \partial_{QE} Q_{\ell+1}^i$$

We know that

$$\mathbb{E}_\ell [\partial_\sigma \partial_{QE} M_{\ell+1} \partial_\sigma (Q_{\ell+1}^i + D_{\ell+1}^i)] = \bar{\xi}_{M,t+1} \frac{\partial_\sigma r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \mathbb{E}_\ell [\bar{M}_{\ell+1} \partial_\sigma (Q_{\ell+1}^i + D_{\ell+1}^i)]$$

so we get

$$\mathbb{E}_\ell [\partial_\sigma \partial_{QE} M_{\ell+1} \partial_\sigma (Q_{\ell+1}^i + D_{\ell+1}^i)] - \partial_\sigma \partial_{QE} M_\ell \partial_\sigma Q_\ell^i \\ = \frac{\partial_\sigma r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \bar{\xi}_{M,t+1} (\mathbb{E}_\ell [\bar{M}_{\ell+1} \partial_\sigma (Q_{\ell+1}^i + D_{\ell+1}^i)] - \bar{M}_\ell \partial_\sigma Q_\ell^i) \\ = \frac{\partial_\sigma r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \bar{\xi}_{M,t+1} \bar{Q}_\ell^i (\partial_\sigma M_\ell - \partial_\sigma M_{\ell+1} \bar{R}_{\ell+1}^{rf})$$

with the last equality coming from

$$\partial_\sigma M_\ell \bar{Q}_\ell^i + \bar{M}_\ell \partial_\sigma Q_\ell^i = \mathbb{E}_\ell \left[\partial_\sigma M_{\ell+1} \left(\bar{Q}_{\ell+1}^i + \bar{D}_{\ell+1}^i \right) + \bar{M}_{\ell+1} \partial_\sigma (Q_{\ell+1}^i + D_{\ell+1}^i) \right].$$

Note that this only depends on i through $\bar{Q}^i$ thus for $\ell > t$

$$\begin{aligned} & \mathbb{E}_\ell \left[\bar{M}_{\ell+1} \partial_{\sigma\sigma} \partial_{QE} Q_{\ell+1}^i - \bar{M}_\ell \partial_{\sigma\sigma} \partial_{QE} Q_\ell^i \right] \\ &= \bar{Q}_\ell^i \mathbb{E}_\ell \left[(\partial_{\sigma\sigma} \partial_{QE} M_\ell) - (\partial_{\sigma\sigma} \partial_{QE} M_{\ell+1}) \bar{R}_{\ell+1}^{rf} - \bar{\xi}_{M,t+1} \bar{M}_\ell \frac{\partial_\sigma r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \frac{\partial_\sigma Q_\ell^1}{\bar{Q}_\ell^1} \right] \end{aligned}$$

where the last term is simplified by noting that $\bar{M}_\ell \frac{\partial_\sigma Q_\ell^1}{\bar{Q}_\ell^1} = \mathbb{E}_\ell \left[\frac{1}{\bar{Q}_\ell^1} \partial_\sigma M_{\ell+1} - \partial_\sigma M_\ell \right]$.

Finally, we have the $\ell = t$ and $s^\ell = s^t$ term which gives

$$\begin{aligned} & \mathbb{E}_t \left[\bar{M}_{t+1} \partial_{\sigma\sigma} \partial_{QE} Q_{t+1}^i - \bar{M}_t \partial_{\sigma\sigma} \partial_{QE} Q_t^i \right] \\ &= \bar{Q}_t^i \mathbb{E}_t \left[(\partial_{\sigma\sigma} \partial_{QE} M_t) - (\partial_{\sigma\sigma} \partial_{QE} M_{t+1}) \bar{R}_{t+1}^{rf} - \frac{\bar{\xi}_{M,t+1} \bar{M}_{t+1}}{1 + \bar{Q}_{t+1}^\infty (s^{t+1})} \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i \right]. \end{aligned}$$

Now we note that all the terms $\bar{M}_{\ell+1} \partial_{\sigma\sigma} \partial_{QE} Q_{\ell+1}^i - \bar{M}_\ell \partial_{\sigma\sigma} \partial_{QE} Q_\ell^i$ in the price effect sum have a component $\bar{Q}_\ell^i \left((\partial_{\sigma\sigma} \partial_{QE} M_\ell) - (\partial_{\sigma\sigma} \partial_{QE} M_{\ell+1}) \bar{R}_{\ell+1}^{rf} \right)$ in them. We gain some tractability by substituting $\partial_{\sigma\sigma} \partial_{QE} M_\ell = \bar{M}_\ell \bar{\xi}_{M,\ell} \partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell) + 2\bar{M}_\ell \partial_\sigma \xi_{M,\ell} \partial_\sigma \partial_{QE} (\tau_\ell Y_\ell)$ and doing so makes

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{\ell \geq 0} \sum_{i \geq 0} \bar{B}_\ell^i (\bar{M}_{\ell+1} \partial_{\sigma\sigma} \partial_{QE} Q_{\ell+1}^i - \bar{M}_\ell \partial_{\sigma\sigma} \partial_{QE} Q_\ell^i) \right] \\ &= \mathbb{E}_0 \left[\sum_{\ell=0}^{\infty} \bar{B}_\ell \bar{M}_\ell \bar{\xi}_{M,\ell} (\partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell) - \partial_{\sigma\sigma} \partial_{QE} (\tau_{\ell+1} Y_{\ell+1})) \right] \\ &\quad - 2\Pr(s^t) \bar{\xi}_{M,t+1} \mathbb{E}_t \left[\sum_{\ell=t+1} \bar{B}_\ell \bar{M}_\ell \frac{\partial_\sigma r_{t+1}^j}{1 + \bar{Q}_{t+1}^\infty} \frac{\partial_\sigma Q_\ell^1}{\bar{Q}_\ell^1} \right] \end{aligned} \tag{32}$$

$$\begin{aligned} &\quad - 2\Pr(s^t) \bar{\xi}_{M,t+1} \mathbb{E}_t \left[\sum_{j \geq 1} \frac{\bar{M}_{t+1}}{1 + \bar{Q}_{t+1}^\infty (s^{t+1})} \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i \right] \\ &\quad + 2\Pr(s^t) \mathbb{E}_t \left[\sum_{\ell=t}^{\infty} \bar{B}_\ell \bar{M}_\ell (\partial_\sigma \xi_{M,\ell} \partial_\sigma \partial_{QE} (\tau_\ell Y_\ell) - \partial_\sigma \xi_{M,\ell+1} \partial_\sigma \partial_{QE} (\tau_{\ell+1} Y_{\ell+1})) \right] \end{aligned} \tag{33}$$

Most of these objects we can easily put some structure on except for

$$\mathbb{E}_0 \left[\sum_{\ell=0}^{\infty} \bar{B}_\ell \bar{M}_\ell \bar{\xi}_{M,\ell} (\partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell) - \partial_{\sigma\sigma} \partial_{QE} (\tau_{\ell+1} Y_{\ell+1})) \right],$$

there we have note that $\overline{B}_\ell = \overline{B}_0 = \overline{B}$, $\overline{M}_\ell = (Q^1)^\ell \overline{M}_0$ and $\overline{\xi}_{M,\ell} = \overline{\xi}_{M,0} = \overline{\xi}_M$. Put together we have

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{\ell=0}^{\infty} \overline{B}_\ell \overline{M}_\ell \overline{\xi}_{M,\ell} (\partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell) - \partial_{\sigma\sigma} \partial_{QE} (\tau_{\ell+1} Y_{\ell+1})) \right] \\ &= \overline{B} \overline{\xi}_M \mathbb{E}_0 \left[\sum_{\ell=0}^{\infty} (Q^1)^\ell (\partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell) - \partial_{\sigma\sigma} \partial_{QE} (\tau_{\ell+1} Y_{\ell+1})) \right] \overline{M}_0 \\ &= \overline{B} \overline{\xi}_M \mathbb{E}_0 \left[\sum_{\ell=0}^{\infty} (Q^1)^\ell (Q^1 - 1) \partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell) \right] \overline{M}_0 \end{aligned}$$

we can then plug into $\partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell)$ to get

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{\ell=0}^{\infty} \overline{B}_\ell \overline{M}_\ell \overline{\xi}_{M,\ell} (\partial_{\sigma\sigma} \partial_{QE} (\tau_\ell Y_\ell) - \partial_{\sigma\sigma} \partial_{QE} (\tau_{\ell+1} Y_{\ell+1})) \right] \\ &= \overline{B} (Q^1 - 1) \overline{\xi}_M \overline{M}_0 \mathbb{E}_0 \left[\sum_{\ell=0}^{\infty} (Q^1)^\ell \sum_{i \geq 0} \partial_{\sigma\sigma} \partial_{QE} Q_\ell^i (\overline{B}_{\ell-1}^i - \overline{B}_\ell^i) \right] \\ &\quad + \overline{B} \overline{\xi}_M \overline{M}_0 (Q^1 - 1) \Pr(s^t) \mathbb{E}_t \left[\sum_{\ell=\ell+1}^{\infty} (Q^1)^\ell \partial_{\sigma\sigma} \left(\frac{r_{t+1}^j(s^{t+1})}{1 + \overline{Q}_{t+1}^\infty} \right) \right] \\ &= \overline{B} (Q^1 - 1) \overline{\xi}_M \frac{\overline{\xi}}{\overline{\xi} - 1} \partial_{\sigma\sigma} \left(\Pr_0(s^t) M_t(s^t) P E_0^j(s^t) \right) \\ &\quad + \overline{B} \overline{\xi}_M \overline{M}_0 (Q^1 - 1) \Pr(s^t) \mathbb{E}_t \left[\sum_{\ell=\ell+1}^{\infty} (Q^1)^\ell \partial_{\sigma\sigma} \left(\frac{r_{t+1}^j(s^{t+1})}{1 + \overline{Q}_{t+1}^\infty} \right) \right] \end{aligned} \tag{34}$$

Going back to the HH version of this perturbation we get

$$\mathbb{E}_t \left[\sum_{\ell=t+1}^{\infty} M_\ell \frac{r_{t+1}^j}{1 + \overline{Q}_{t+1}^\infty} \right] = 0$$

As second order expansion of this gives

$$\overline{M}_0 \mathbb{E}_t \left[\sum_{\ell=t+1}^{\infty} (Q^1)^\ell \partial_{\sigma\sigma} \left(\frac{r_{t+1}^j(s^{t+1})}{1 + \overline{Q}_{t+1}^\infty} \right) \right] = -2 \mathbb{E}_t \left[\sum_{\ell=t+1}^{\infty} \partial_\sigma M_\ell \frac{\partial_\sigma r_{t+1}^j(s^{t+1})}{1 + \overline{Q}_{t+1}^\infty} \right] \tag{35}$$

Putting all together we get (combining equations (29),(32),(34), and (35))

$$\begin{aligned}
\left(\frac{\bar{\xi}}{\bar{\xi}-1} \right) \partial_{\sigma\sigma} P E_{j,t,\epsilon} &= \frac{-2\bar{\xi}_M}{(1-\bar{B}_0(Q^1-1)) (Q^1)^t} \mathbb{E}_t \left[\sum_{\ell=t+1} \bar{B}_\ell \frac{\partial_\sigma r_{t+1}^j}{1+\bar{Q}_{t+1}^\infty} \frac{\partial_\sigma Q_t^1}{\bar{Q}_t^1} \right] \\
&\quad - \frac{2\bar{B}\bar{\xi}_M(Q^1-1)}{(1-\bar{B}(Q^1-1))} \mathbb{E}_t \left[\sum_{t=t+1}^\infty (Q^1)^{t-t} \partial_\sigma \ln M_t \frac{\partial_\sigma r_{t+1}^j(s^{t+1})}{1+\bar{Q}_{t+1}^\infty} \right] \\
&\quad \frac{-2\bar{\xi}_M}{(1-\bar{B}(Q^1-1)) (Q^1)^t} \mathbb{E}_t \left[\sum_{j \geq 1} \frac{Q^1}{1+\bar{Q}_{t+1}^\infty(s^{t+1})} \partial_\sigma r_{t+1}^j \partial_\sigma r_{t+1}^i \right] \\
&\quad - \frac{2\bar{B}}{(1-\bar{B}(Q^1-1))} \mathbb{E}_t \left[\sum_{t=t}^\infty (Q^1)^{t-t} \frac{\partial_\sigma r_{t+1}^j(s^{t+1})}{1+\bar{Q}_{t+1}^\infty} (\partial_\sigma \xi_{M,t} - \partial_\sigma \xi_{M,t+1}) \right]
\end{aligned}$$

as desired.